Aphanics and Creative Curvature: A Geometric Theory of Hidden Generation in Dialectical Cognitive Systems

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Abstract

Cognitive and generative systems routinely face the problem of distinguishing genuine creativity from random displacement in state space. Without a formal distinction, controllers cannot tell whether a system is refining, hallucinating, pivoting, or genuinely creating. We introduce a geometric framework, which we term *aphanics*, for systems possessing *hidden generators*: nonzero elements annihilated by the system's own internal action operator, $a(p) \in \ker(L_p) \setminus \{0\}$.

Our main results are geometric and exact. In the sedenion algebra the locus of such observers is a smooth manifold, an $\mathbb{R}^+$-cone over the homogeneous space $G_2/\mathrm{SU}(2)$, over which the kernel assignment defines a rank-4 vector bundle with a canonical metric connection. The Berry curvature of this connection admits a closed polynomial form $\mathcal{C}_{a,b}(p) = \pi_p(L_b\, G_p^+\, L_a - L_a\, G_p^+\, L_b)\,\pi_p$, bypassing eigenprojector contour integrals (Theorem 3.9), and its holonomy algebra is exactly $\mathfrak{u}(2) = \mathfrak{u}(1) \oplus \mathfrak{su}(2)$ (Theorem 4.13, Proposition 4.15), with the central complex structure realized by right multiplication by the Cayley–Dickson doubling generator. All basepoint computations decompose with integer coefficients onto 't Hooft self-dual tensors, and are certified in exact arithmetic: both the Q-centralizer dimension and the basepoint curvature-span decomposition are exact, the latter with identically zero residual and exact integer coefficients.

Building on this geometry we *propose* an operational layer for monitoring dialectical inference systems: a creative-curvature scalar κ_γ, a phase-transition score Φ_k, a survival-gated verified creativity score C_k^{surv}, a seven-regime diagnostic taxonomy, and a U(1) holonomy phase $\theta(\Gamma)$ over closed dialectical arcs that certifies net structural turning. The geometric results (§3–§4) are theorems; the monitoring layer (§5 onward) is a framework proposal whose empirical validation in any deployed system is future work.

Keywords: aphanics, creative curvature, dialectical inference, Berry curvature, holonomy, geometric phase, hidden-sector dynamics, Cayley–Dickson algebras, cognitive systems.

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1 Introduction

1.1 The Problem: Creativity Is Not Displacement

Cognitive and generative systems produce novel outputs, but novelty alone does not distinguish genuine creativity from hallucination or random drift. Existing metrics (perplexity, compression gain, distributional divergence) measure displacement in observation space, not structural transformation in the system's internal geometry. A system can move far in state space without turning. It can turn without moving. It can move and turn while violating every constraint it was built to respect. None of these is creativity. Creativity is turning in a way that deforms the constraint manifold itself while remaining on it throughout.

The analogy is precise: in the Ryōan-ji rock garden in Kyoto, fifteen stones are arranged so that from any standing position exactly fourteen are visible; the fifteenth is hidden by the geometry of the arrangement, not by accident or deficiency. Remove it and the composition collapses.

This paper formalizes the hidden stone. In any system with an internal action operator L_p, there may exist generators a such that $L_p \cdot a = 0$ while $a \neq 0$: elements that are annihilated by the system's own action yet remain structurally present. We call these *aphanes generators*, from the Greek *aphanés* ($\alpha\phi\alpha\nu\acute{\varepsilon}\varsigma$): the non-apparent. The study of their dynamics, geometry, and operational consequences constitutes the framework we present here.

The central interpretive move is a sign-flip: the kernel of L_p is conventionally treated as a degeneracy, a locus where the operator fails. We reinterpret it as the locus of maximal structural richness. The hidden generator is not what the system cannot do; it is what prevents the system from collapsing into mechanical repetition. This reinterpretation has concrete mathematical content: the hidden sector carries bundle structure, curvature, and holonomy that provide a geometric diagnostic for structural turning.

1.2 Contributions

This paper makes six contributions, in two tiers. Contributions 1–2 are *theorems*, proved in the sedenion model (Sections 3–4). Contributions 3–6 are a *framework proposal*: an instrumentation layer for dialectical systems whose validity in any deployed system requires that system to expose a hidden-sector bundle with the stated structure, and whose empirical validation is future work.

1. **The aphanic Berry bundle** (Section 3; *theorem*): the kernel assignment $p \mapsto \ker L_p$ over the aphanic locus, realized as a smooth rank-4 vector bundle with a kernel-projector connection, whose Berry curvature is identified in closed form with the algebraic curvature operator $\mathcal{C}_{a,b}(p)$.

2. **The algebraic structure of aphanic curvature** (Section 4; *theorem*): a closed-

form pseudoinverse and projector, G_2-equivariance, the Q-centralizer vanishing theorem, and the rank theorem computing the curvature span, capped (Proposition 4.15) to the holonomy algebra $\mathfrak{u}(2) = \mathfrak{u}(1) \oplus \mathfrak{su}(2)$, and sharpened by an eigenspace-mixing trichotomy that counts the rank.

3. **Creative curvature and discrete operational formulas** (Section 5): the creative-curvature scalar κ_γ, its discrete turn-angle and G-metric forms, exogenous subtraction procedures, and the phase transition score Φ_k.

4. **The survival-gated creativity score** C_k^{surv} (Section 7): a triple product of curvature, verification weight, and survival horizon that separates five distinct failure modes.

5. **A seven-regime operational taxonomy** (Section 8): a classifier with hysteresis that maps continuous curvature-verification trajectories to discrete operational states.

6. **The holonomy phase** $\theta(\Gamma)$ (Section 9; *proposal*): a U(1) geometric holonomy phase over closed dialectical arcs that certifies net structural turning and distinguishes turning loops from drift. It is geometric, not topological: the aphanic base is simply connected, so $\theta(\Gamma)$ is curvature-driven and deformation-sensitive, not a homotopy invariant.

1.3 Scope and Deliberate Omissions

This paper operates at the *abstract geometric level*. We define the aphanes generator via the kernel of a left-multiplication operator L_p, derive curvature and a holonomy phase from connection geometry on the resulting bundle, and establish operational formulas for monitoring dialectical systems. The framework is motivated at this abstract level; its v6 core (Sections 3–4) instantiates the construction concretely in the sedenion algebra $\mathbb{S}$, while the companion algebra carries the same construction up the Cayley–Dickson tower. The downstream operational formulas (Sections 5–9) depend only on the existence and connection geometry of the hidden sector, and may be read without committing to a unique implementation algebra or inference architecture. This separation is deliberate.

The mathematical foundations of aphanics (the characterization of $\ker(L_p)$, its dimension, spectral structure, and the topology of the associated bundle) are developed in a companion paper, *Algebraic Hiddenness in the Sedenion Algebra* [21]. The four foundational statements the geometric core actually uses (the Two-Condition characterization of $\ker(L_p)$, its dimension $\dim K_p = 4$, the 4–8–4 spectrum of G_p, and the identification of the unit aphanic locus with $G_2/\mathrm{SU}(2)$) are stated where they are used in Sections 3–4 and cited to that companion, which is published and publicly available [21]. The present paper is not self-contained on its foundations; every foundational statement it uses is stated, proved, and independently auditable there. Section 4.1 provides a brief sketch of the motivating algebraic setting.

1.4 Fixed definitions

Although the foundations are the companion's, the three terms this paper uses inside formal statements are fixed here, before first such use, in the companion's own formulation [21].

Definition 1.1 (Aphanic element)**.** Let $p = (x, y) \in \mathbb{S}$ under the Cayley–Dickson identification $\mathbb{S} = \mathbb{O} \oplus \mathbb{O}$. Then p is *aphanic* if $x, y \in \mathrm{Im}(\mathbb{O})$ and $\|x\| = \|y\| \neq 0$, $\langle x, y \rangle = 0$, where $\langle \cdot, \cdot \rangle$ and $\| \cdot \|$ are the standard Euclidean inner product and norm on $\mathbb{O} \cong \mathbb{R}^8$.

Definition 1.2 (Aphanic locus)**.** The *aphanic locus* is the set of all aphanic elements of $\mathbb{S}$; its unit normalization is the homogeneous space $G_2/\mathrm{SU}(2)$ used throughout this paper.

Definition 1.3 (Aphanic twist)**.** The *aphanic twist* of the kernel bundle over the unit aphanic locus is its secondary characteristic class in $H^6(G_2/\mathrm{SU}(2); \mathbb{Z}/2) \cong \mathbb{Z}/2$.

Remark 1.4. Definition 1.3 names an invariant, not a phenomenon: the twist exists as a class whatever its value, its nonvanishing is a theorem of the companion, and the definition deliberately does not identify the class with a holonomy character. The class classifies the bundle and is detected by five- and six-dimensional cycles; the holonomy phase $\theta(\Gamma)$ of this paper is a connection-dependent quantity on arcs, and the two are never interchanged.

1.5 Roadmap

Section 2 surveys related work across information geometry, geometric deep learning, creativity metrics, and topological methods. Section 3 constructs the aphanic Berry bundle and identifies its Berry curvature with a closed-form algebraic operator. Section 4 establishes the algebraic structure of that curvature and proves the $\mathfrak{u}(2)$ rank theorem. Section 5 provides discrete formulas for the creative-curvature scalar. Section 6 specifies the state vector. Section 7 defines verified creativity. Section 8 presents the regime taxonomy. Section 9 establishes the holonomy phase. Section 10 discusses implications, limitations, and connections. Section 11 concludes, and Appendix C tabulates the explicit basepoint data underlying the rank theorem.

2 Related Work

2.1 Information Geometry and Natural Gradient

The idea that learning and inference have intrinsic geometric structure traces to Amari's information geometry [2], where the Fisher information matrix endows parameter space with a Riemannian metric and the natural gradient [1] replaces Euclidean descent with

geodesic motion on the statistical manifold. Martens [15] extended this to practical deep learning via K-FAC approximations. These methods measure curvature of the *loss landscape*: how predictions change as parameters vary. Aphanics measures something different: curvature of the *hidden sector*: how the structurally invisible generators rotate as the system moves. The Fisher metric sees everything; the aphanes metric sees only what is hidden.

2.2 Geometric Deep Learning

Bronstein et al. [6] unified CNNs, GNNs, and Transformers under the umbrella of geometric deep learning: architectures that respect the symmetry group of their input domain. Equivariant networks [8, 23] preserve group structure through layers. This work focuses on *visible* symmetry: the group actions that transform inputs and outputs. Aphanics addresses the complementary question: what structure is present but invisible to the system's own measurement operator? The hidden sector is not a latent space in the variational autoencoder sense [14]; it is a kernel of an algebraic action, not a learned compression.

2.3 Creativity Metrics

Boden [5] distinguished combinational, exploratory, and transformational creativity, arguing that the third requires restructuring the conceptual space itself. Schmidhuber [18] formalized creativity as compression progress: a system is creative when it finds shorter descriptions of its experience. Wiggins [24] gave a formal framework for creative systems based on search in conceptual spaces. These approaches measure *what* is produced. Aphanics measures *how the system turns while producing it*: the geometry of the trajectory, not the novelty of the output. Creative curvature κ_γ captures Boden's transformational creativity geometrically: the constraint manifold deforms under the trajectory.

2.4 Topological Methods in Machine Learning

Persistent homology [7, 10] and topological data analysis extract shape invariants from data. Rieck et al. [17] applied persistence to neural network loss surfaces. Naitzat et al. [16] tracked topological changes through layers. These methods study the topology of data or loss landscapes. The holonomy phase $\theta(\Gamma)$ in this paper is a geometric invariant of a *trajectory*, a closed loop in the space of system states, not of the data itself. It measures irreversible structural passage, not homological features of a point cloud.

2.5 Dialectical and Self-Corrective Architectures

Thesis-antithesis-synthesis as a computational paradigm has antecedents in argumentation frameworks [9], debate-based AI alignment [12], and self-play systems [19]. Constitutional

AI [3] uses self-critique to improve outputs. Friston's free-energy principle [11] provides a complementary perspective: systems minimize surprise by maintaining generative models, and the boundary between model and environment is itself a structural object. The present formalism is motivated by dialectical self-correction architectures in which each inference step generates a thesis and antithesis before synthesizing. The aphanics framework provides the geometric infrastructure for monitoring such systems: κ_γ tracks whether the dialectical process is genuinely turning or merely oscillating, and $\theta(\Gamma)$ determines whether a completed arc has produced irreversible structural change.

2.6 Geometric Phases and Holonomy

The holonomy phase $\theta(\Gamma)$ defined in Section 9 is a geometric phase in the sense of Berry [4]: the holonomy of a connection on a parameter-dependent eigenspace bundle around a closed loop. Simon [20] identified Berry's phase as the holonomy of a natural connection on a line bundle over parameter space, and Wilczek and Zee [25] generalized it to non-abelian gauge structure when the eigenspace is degenerate. In the algebraic setting motivating this work, the rank-4 kernel bundle admits a canonical connection whose holonomy algebra is exactly $\mathfrak{u}(2)$ (Theorem 4.13, Proposition 4.15); the holonomy phase $\theta(\Gamma)$ used here is its abelian U(1) component, while the SU(2) component is the Wilczek–Zee generalization and is not used in this paper. Separately, the bundle carries a genuine *topological* invariant, a nonzero $\mathbb{Z}/2$ characteristic class (the "aphanic twist") in $H^6(G_2/\mathrm{SU}(2); \mathbb{Z}/2)$, which classifies the bundle itself and is unrelated to the loop phase (the base is simply connected). The U(1) generator admits an explicit ambient-operator description: it is the restriction to the kernel of right multiplication by the doubling generator of the algebra. This identification is tower-universal across Cayley–Dickson levels $n = 2^k \geq 16$ [21], so the underlying *algebraic* identity is not a feature of the sedenion level alone but an inheritance from the Cayley–Dickson doubling structure itself. We claim no holonomy statement above $n = 16$: the bundle, connection and curvature used here are constructed only at the sedenion level. The framework presented here applies geometric phase theory to cognitive systems rather than quantum mechanics: the "parameter" is the system state, the "eigenspace" is the hidden sector $\ker(L_p)$, and the holonomy classifies irreversible structural transformation rather than adiabatic evolution.

3 The Aphanic Berry Bundle

This section constructs the geometric object whose holonomy algebra is computed in §4. The aphanic locus $\mathcal{P}_{\mathrm{aph}} \subset \mathbb{S}$ is a smooth real-analytic manifold; the assignment $p \mapsto K_p = \ker L_p$ defines a smooth real rank-4 vector bundle $K \to \mathcal{P}_{\mathrm{aph}}$; and the kernel-projector prescription $\nabla := \pi \circ d \circ \pi$ defines a metric connection on K. The principal result of the section (Theorem 3.9) identifies the resulting Berry curvature 2-form, restricted to fibers, with the closed-form curvature operator $\mathcal{C}_{a,b}(p)$ used throughout §4.

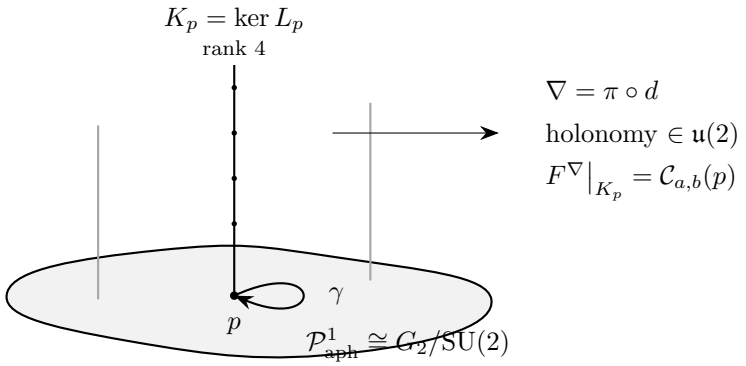


Figure 1: The aphanic Berry bundle. Over each point p of the unit aphanic locus $\mathcal{P}^1_{\text{aph}} \cong G_2/\mathrm{SU}(2)$ sits the rank-4 kernel fiber $K_p = \ker L_p$. The kernel-projector connection $\nabla = \pi \circ d$ (Definition 3.5) has holonomy $\mathfrak{u}(2)$, and its curvature 2-form restricted to the fibers coincides with the closed-form operator $\mathcal{C}_{a,b}(p)$ (Theorem 3.9).

The proof of the bridge theorem is a single application of eigenprojector perturbation theory [13], made transparent by the closed-form representation of π_p in $\mathbb{R}[G_p]$ (Proposition 4.2, restated below) and three algebraic identities that follow from the universal 4–8–4 spectrum.

3.1 The Aphanic Locus as a Smooth Manifold

We retain the notation of §4: $\mathbb{S} = \mathbb{O} \oplus \mathbb{O}\,\ell$ with the canonical CD product, embeddings $\iota_1, \iota_2 \colon \mathbb{O} \to \mathbb{S}$, and the Two-Condition Theorem characterizing the aphanic locus.

Proposition 3.1 (Aphanic locus is a real-analytic manifold)**.** *The aphanic locus $\mathcal{P}_{\text{aph}} \subset \mathbb{S}$, identified with $\{(x, y) \in \mathrm{Im}(\mathbb{O})^{\oplus 2} : \|x\| = \|y\| > 0,\ \langle x, y\rangle = 0\}$ via $p = \iota_1(x) + \iota_2(y)$, is a real-analytic submanifold of $\mathbb{S}$ of dimension 12. The unit slice $\mathcal{P}^1_{\text{aph}} := \{p \in \mathcal{P}_{\text{aph}} : \|p\|^2 = 2\}$ is an 11-dimensional smooth submanifold, a single G_2-orbit diffeomorphic to $G_2/\mathrm{SU}(2)$ [21], and $\mathcal{P}_{\text{aph}}$ is the $\mathbb{R}^+$-cone over $\mathcal{P}^1_{\text{aph}}$.*

Proof. The two real-analytic constraints $f_1(x, y) := \|x\|^2 - \|y\|^2$ and $f_2(x, y) := \langle x, y\rangle$ on $\mathrm{Im}(\mathbb{O})^{\oplus 2} \cong \mathbb{R}^{14}$ have differentials $df_1 = 2(x\,dx - y\,dy)$ and $df_2 = y\,dx + x\,dy$. At any (x, y) with x, y both nonzero and orthogonal, the gradients are linearly independent: the gradient of f_1 involves the components $(x, -y)$ in the (dx, dy) basis, while the gradient of f_2 involves (y, x). Linear dependence would force $x \parallel y$ as vectors in $\mathrm{Im}(\mathbb{O})$, contradicting $\langle x, y\rangle = 0$ with $\|x\| = \|y\| > 0$. The regular-value theorem applies, giving codimension 2 and dimension $14 - 2 = 12$. The unit slice is cut out by the additional smooth constraint $\|x\|^2 + \|y\|^2 = 2$, which is independent of the previous two for the same reason; this is dimension 11, and matches the homogeneous-space description $G_2/\mathrm{SU}(2)$ from the companion [21]. □

Remark 3.2. The closed-form expressions for π_p and G_p^+ depend only on G_p and $\|p\|^2$, so the bridge identity proven in Theorem 3.9 is homogeneous of weight -2 under radial

scaling. Working on $\mathcal{P}_{\text{aph}}$ rather than its unit slice is therefore harmless and convenient.

3.2 The Kernel Bundle

For each $p \in \mathcal{P}_{\text{aph}}$, the fiber $K_p := \ker L_p$ is a 4-dimensional real subspace of $\mathbb{S}$ [21]. The total space

$$K \;:=\; \{\, (p, v) \in \mathcal{P}_{\text{aph}} \times \mathbb{S} \;:\; v \in K_p \,\} \tag{1}$$

projects onto $\mathcal{P}_{\text{aph}}$ by $(p, v) \mapsto p$.

Proposition 3.3 (Kernel bundle smoothness)**.** *The projection $K \to \mathcal{P}_{\text{aph}}$ is a smooth real rank-4 vector bundle, and the family of orthogonal projectors $p \mapsto \pi_p$ is real-analytic on $\mathcal{P}_{\text{aph}}$.*

Proof. The closed-form representation of π_p from Proposition 4.2 of §4.2,

$$\pi_p \;=\; I \;-\; \frac{3}{2\|p\|^2}\, G_p \;+\; \frac{1}{2\|p\|^4}\, G_p^2, \tag{2}$$

is a polynomial in $G_p = L_p^\top L_p$ with coefficients that are real-analytic in $\|p\|^2$ on $\mathcal{P}_{\text{aph}}$ (where $\|p\|^2 > 0$). Since $p \mapsto L_p$ is linear in p, the matrix entries of G_p are quadratic polynomials in the components of p, and (2) is real-analytic on $\mathcal{P}_{\text{aph}}$.

Smoothness of π_p as a projector field upgrades the kernel assignment $p \mapsto K_p$ to a smooth subbundle: locally on a neighborhood U of any $p_* \in \mathcal{P}_{\text{aph}}$, choose four vectors $v_1, \ldots, v_4 \in K_{p_*}$ and set $w_k(p) := \pi_p\, v_k$ for $p \in U$. By continuity, $\{w_k(p)\}_{k=1}^4$ remains linearly independent on a sufficiently small neighborhood, providing a smooth local frame for $K|_U$. □

Remark 3.4 (Closed form vs. contour-integral construction). The standard route to bundle smoothness in the eigenprojector literature constructs π_p via a Riesz contour integral of the resolvent $(zI - G_p)^{-1}$ on a contour separating $\ker G_p$ from the rest of the spectrum [13]. The polynomial representation (2) bypasses the contour entirely; smoothness is read off algebraically. This is the same advantage exploited throughout §4: every assertion that would ordinarily require eigendecomposition is instead a polynomial identity in G_p.

3.3 The Kernel-Projector Connection

The bundle K inherits a fiber inner product from $\mathbb{S}$: for $u, v \in K_p$, $\langle u, v\rangle_K := \langle u, v\rangle_{\mathbb{S}}$.

Definition 3.5 (Kernel-projector connection)**.** The *kernel-projector connection* on K is

$$\nabla \;:=\; \pi \circ d, \qquad (\nabla_X s)(p) \;:=\; \pi_p\,(X[s])(p), \tag{3}$$

for $X \in T_p\mathcal{P}_{\text{aph}}$ and $s\colon \mathcal{P}_{\text{aph}} \to \mathbb{S}$ a section of K (regarded as an $\mathbb{S}$-valued function satisfying $\pi_p s(p) = s(p)$).

Lemma 3.6 (Metric compatibility)**.** *For sections s, t of K and tangent vector X, $X\langle s,t\rangle_K = \langle \nabla_X s, t\rangle_K + \langle s, \nabla_X t\rangle_K$.*

Proof. Since $s = \pi_p s$ and $t = \pi_p t$, and $\pi_p^\top = \pi_p$,

$$\langle \nabla_X s, t\rangle + \langle s, \nabla_X t\rangle \;=\; \langle \pi_p X[s], t\rangle + \langle s, \pi_p X[t]\rangle \;=\; \langle X[s], t\rangle + \langle s, X[t]\rangle \;=\; X\langle s,t\rangle. \;\square$$

The connection ∇ is the standard kernel-projector (Berry) connection of a smoothly varying subspace of a fixed inner-product space [4, 13].

3.4 The Berry Curvature 2-Form

The curvature of ∇ is the $\mathrm{End}(K)$-valued 2-form $F^\nabla := \nabla \circ \nabla \in \Omega^2(\mathcal{P}_{\mathrm{aph}}; \mathrm{End}(K))$.

Lemma 3.7 (Closed-form expression for F^∇)**.** *Let $\partial_a, \partial_b \in T_p\mathcal{P}_{\mathrm{aph}}$. Then*

$$F^\nabla_{a,b}(p)\big|_{K_p} \;=\; \pi_p \left[\, \partial_a \pi_p,\; \partial_b \pi_p \,\right] \pi_p, \tag{4}$$

where $[\cdot,\cdot]$ is the operator commutator on $\mathrm{End}(\mathbb{S})$.

Proof. Acting on a section s of K, $\nabla_a s = \pi_p\, \partial_a s$. Thus

$$\nabla_a \nabla_b s \;=\; \pi_p\, \partial_a (\pi_p\, \partial_b s) \;=\; \pi_p\, (\partial_a \pi_p)\, (\partial_b s) \;+\; \pi_p\, \pi_p\, \partial_a \partial_b s.$$

The torsion-free curvature is the antisymmetrization in (a,b):

$$F^\nabla_{a,b} s \;=\; \nabla_a \nabla_b s - \nabla_b \nabla_a s \;=\; \pi_p \left[(\partial_a \pi_p)(\partial_b s) - (\partial_b \pi_p)(\partial_a s) \right],$$

the second-derivative terms cancelling by Schwarz. Now $s = \pi_p\, s$, so $\partial_b s = (\partial_b \pi_p)\, s + \pi_p\, \partial_b s$. Substituting and using the identity $\pi_p\, (\partial_a \pi_p)\, \pi_p = 0$ (Remark 3.8, below), the second piece vanishes and

$$F^\nabla_{a,b}\, s \;=\; \pi_p \left[(\partial_a \pi_p)(\partial_b \pi_p) - (\partial_b \pi_p)(\partial_a \pi_p) \right] s,$$

which is (4). $\square$

Remark 3.8 (Identity $\pi_p(\partial \pi_p)\pi_p = 0$). Differentiating $\pi_p^2 = \pi_p$ along any tangent direction a gives $(\partial_a \pi_p)\, \pi_p + \pi_p\, (\partial_a \pi_p) = \partial_a \pi_p$. Sandwiching with π_p on both sides yields $2\, \pi_p\, (\partial_a \pi_p)\, \pi_p = \pi_p\, (\partial_a \pi_p)\, \pi_p$, hence $\pi_p\, (\partial_a \pi_p)\, \pi_p = 0$. Equivalently, $\partial_a \pi_p$ is purely off-diagonal with respect to the decomposition $\mathbb{S} = K_p \oplus K_p^\perp$.

3.5 The Closed-Form Bridge

We now state and prove the principal theorem of §3: the Berry curvature, restricted to fibers of K, equals the closed-form operator $\mathcal{C}_{a,b}(p)$ used throughout §4.

Theorem 3.9 (Closed-form bridge)**.** *For every $p \in \mathcal{P}_{\mathrm{aph}}$ and every pair of tangent directions $a, b \in T_p\mathcal{P}_{\mathrm{aph}} \subset \mathrm{Im}(\mathbb{S})$ (the extension of the right-hand side to all of $\mathrm{Im}(\mathbb{S})$ is harmless by Corollary 4.11 of §4, which shows it vanishes on the normal directions),*

$$F^{\nabla}_{a,b}(p)\big|_{K_p} \;=\; \mathcal{C}_{a,b}(p) \;=\; \pi_p \left(L_b\, G_p^{+}\, L_a \;-\; L_a\, G_p^{+}\, L_b\right) \pi_p. \tag{5}$$

Proof. By Lemma 3.7,

$$F^{\nabla}_{a,b}(p)\big|_{K_p} \;=\; \pi_p \,(\partial_a \pi_p)(\partial_b \pi_p)\, \pi_p \;-\; \pi_p \,(\partial_b \pi_p)(\partial_a \pi_p)\, \pi_p.$$

We compute $\pi_p \,(\partial_a \pi_p)(\partial_b \pi_p)\, \pi_p$ explicitly, then antisymmetrize in (a, b).

Step 1 — Eigenprojector perturbation. The standard formula for the derivative of an isolated spectral projector [13, Ch. II §1.4], applied to the projector π_p onto the zero eigenspace of G_p, reads

$$\partial_a \pi_p \;=\; -\,\pi_p \,(\partial_a G_p)\, G_p^{+} \;-\; G_p^{+}\,(\partial_a G_p)\, \pi_p. \tag{6}$$

Derivation: differentiating $G_p\, \pi_p = 0$ gives $(\partial_a G_p)\, \pi_p + G_p\,(\partial_a \pi_p) = 0$, hence $G_p\,(\partial_a \pi_p) = -(\partial_a G_p)\, \pi_p$. Multiplying on the left by G_p^{+} and using $G_p^{+}\, G_p = I - \pi_p$ projects out the off-diagonal block $(I - \pi_p)\,(\partial_a \pi_p)\, \pi_p = -G_p^{+}\,(\partial_a G_p)\, \pi_p$. The transposed identity $\pi_p\, G_p = 0$ gives the other off-diagonal block $\pi_p\,(\partial_a \pi_p)\,(I - \pi_p) = -\pi_p\,(\partial_a G_p)\, G_p^{+}$. Summing, and using $\pi_p\,(\partial_a \pi_p)\, \pi_p = 0$ (Remark 3.8), gives (6).

The directional derivative of $G_p = L_p^{\top} L_p$ is

$$\partial_a G_p \;=\; L_a^{\top} L_p \;+\; L_p^{\top} L_a \;=\; -(L_a L_p + L_p L_a) \;=\; -\,\{L_a, L_p\}, \tag{7}$$

where the second equality uses $L_p^{\top} = -L_p$, $L_a^{\top} = -L_a$ for imaginary a, p.

Step 2 — Sandwich identities. The two terms in (6) are killed asymmetrically by left- and right-multiplication by π_p. From $\pi_p\, G_p^{+} = G_p^{+}\, \pi_p = 0$ (Proposition 4.2),

$$\pi_p\,(\partial_a \pi_p) \;=\; -\,\pi_p\,(\partial_a G_p)\, G_p^{+}, \qquad (\partial_b \pi_p)\, \pi_p \;=\; -\,G_p^{+}\,(\partial_b G_p)\, \pi_p. \tag{8}$$

Sandwiching the product gives

$$\begin{aligned} \pi_p \,(\partial_a \pi_p)(\partial_b \pi_p)\, \pi_p \;&=\; \big(\pi_p\,(\partial_a \pi_p)\big)\,\big((\partial_b \pi_p)\, \pi_p\big) \\ &=\; \pi_p\,(\partial_a G_p)\,(G_p^{+})^2\,(\partial_b G_p)\, \pi_p \end{aligned} \tag{9}$$

(the two minus signs from (8) cancel).

Step 3 — Four cross-terms; three vanish. Substituting $\partial_a G_p = -\{L_a, L_p\}$ from (7) into (9) and expanding the product of two anticommutators gives four terms (the two

minus signs again cancelling):

$$\pi_p \{L_a, L_p\} (G_p^+)^2 \{L_b, L_p\} \pi_p \;=\; \pi_p \left[T_1 + T_2 + T_3 + T_4 \right] \pi_p, \tag{10}$$

where

$$T_1 = L_a L_p (G_p^+)^2 L_b L_p, \quad T_2 = L_a L_p (G_p^+)^2 L_p L_b, \quad T_3 = L_p L_a (G_p^+)^2 L_b L_p, \quad T_4 = L_p L_a (G_p^+)^2 L_p L_b.$$

The boundary identity

$$L_p \pi_p \;=\; \pi_p L_p \;=\; 0 \tag{11}$$

holds because π_p projects onto $\ker L_p$ (giving $L_p \pi_p = 0$) and $\pi_p L_p = (L_p^\top \pi_p^\top)^\top = (-L_p \pi_p)^\top = 0$ for imaginary p. By inspection of the four terms in (10):

- $\pi_p T_1 \pi_p = 0$: rightmost factor $L_p \pi_p$ vanishes.
- $\pi_p T_3 \pi_p = 0$: leftmost factor $\pi_p L_p$ vanishes.
- $\pi_p T_4 \pi_p = 0$: both outer factors vanish (either suffices).
- $\pi_p T_2 \pi_p$: the two interior L_p factors flank $(G_p^+)^2$.

Only T_2 survives.

Step 4 — Polynomial collapse of T_2. Since $(G_p^+)^2 \in \mathbb{R}[G_p]$ by Proposition 4.2 and L_p commutes with G_p (because $G_p = -L_p^2$ for imaginary p, and L_p commutes with L_p^2), L_p commutes with $(G_p^+)^2$. Hence

$$L_p (G_p^+)^2 L_p \;=\; (G_p^+)^2 L_p^2 \;=\; -(G_p^+)^2 G_p.$$

The pseudoinverse identity $G_p^+ G_p = I - \pi_p$, combined with $G_p^+ \pi_p = 0$, yields

$$(G_p^+)^2 G_p \;=\; G_p^+ (G_p^+ G_p) \;=\; G_p^+ (I - \pi_p) \;=\; G_p^+, \tag{12}$$

the polynomial identity advertised at the start of the proof. Substituting,

$$L_p (G_p^+)^2 L_p \;=\; -G_p^+, \qquad \text{hence} \qquad T_2 \;=\; L_a \left(L_p (G_p^+)^2 L_p \right) L_b \;=\; -\, L_a G_p^+ L_b.$$

Sandwiching with π_p,

$$\pi_p (\partial_a \pi_p)(\partial_b \pi_p) \pi_p \;=\; \pi_p T_2 \pi_p \;=\; -\, \pi_p L_a G_p^+ L_b \pi_p. \tag{13}$$

Step 5 — Antisymmetrize. From (13),

$$\begin{aligned} F^{\nabla}_{a,b}(p)\big|_{K_p} &= \pi_p\,(\partial_a\pi_p)(\partial_b\pi_p)\,\pi_p \;-\; \pi_p\,(\partial_b\pi_p)(\partial_a\pi_p)\,\pi_p \\ &= -\,\pi_p\,L_a\,G_p^{+}\,L_b\,\pi_p \;+\; \pi_p\,L_b\,G_p^{+}\,L_a\,\pi_p \\ &= \pi_p\,(L_b\,G_p^{+}\,L_a - L_a\,G_p^{+}\,L_b)\,\pi_p \;=\; \mathcal{C}_{a,b}(p), \end{aligned}$$

which is (5). □

Remark 3.10 (Numerical verification). Theorem 3.9 has been verified numerically by central finite-difference perturbation of π_p at 50 random aphanic samples, comparing

$$F^{\mathrm{FD}}_{a,b}(p) \;:=\; \pi_p\,[\delta_a\pi,\,\delta_b\pi]\,\pi_p, \qquad \delta_a\pi := \tfrac{1}{2\epsilon}(\pi_{p+\epsilon a} - \pi_{p-\epsilon a}),$$

to the closed-form $\mathcal{C}_{a,b}(p)$. The discrepancy $\|F^{\mathrm{FD}} - \mathcal{C}\|_F$ scales as ϵ^2 over four decades $\epsilon \in [10^{-6}, 10^{-3}]$, with floor at $\epsilon \approx 10^{-5}$ where the residual is 1.1×10^{-10}, dominated by eigensolver roundoff. The opposite-sign hypothesis $\|F^{\mathrm{FD}} + \mathcal{C}\|_F$ remains $O(\|\mathcal{C}\|_F)$ at every ϵ, distinguishing the sign by nine orders of magnitude. The verification is reproducible from the probe `op_F2_berry_kernel_match.cpp` in the supplementary archive.

3.6 Summary

The aphanic locus $\mathcal{P}_{\mathrm{aph}}$ is a smooth real-analytic manifold (Proposition 3.1). The kernel assignment $p \mapsto K_p$ defines a smooth real rank-4 vector bundle $K \to \mathcal{P}_{\mathrm{aph}}$ whose fiber projector π_p is real-analytic in p (Proposition 3.3); the latter follows from the closed-form polynomial expression of π_p in G_p rather than from any abstract eigenprojector argument. The kernel-projector connection $\nabla = \pi \circ d$ is the natural metric connection (Lemma 3.6); its curvature 2-form F^{∇}, restricted to fibers, has the closed-form expression

$$F^{\nabla}_{a,b}(p)\big|_{K_p} \;=\; \pi_p\,(L_b\,G_p^{+}\,L_a - L_a\,G_p^{+}\,L_b)\,\pi_p \;=\; \mathcal{C}_{a,b}(p),$$

the principal output of the section (Theorem 3.9). This identity is the geometric foundation of §4: the linear span $\mathcal{F}(p) = \mathrm{span}_{\mathbb{R}}\{\mathcal{C}_{a,b}(p)\}$, computed there as a purely algebraic object, is the pointwise curvature span of the aphanic Berry bundle. The rank theorem of §4.6 shows this span is $\mathfrak{u}(2)$; the holonomy-cap proposition (Proposition 4.15) then upgrades the pointwise span to the full holonomy algebra of the kernel-projector connection, $\mathfrak{u}(2)$ at every point of $\mathcal{P}_{\mathrm{aph}}$.

4 The Algebraic Structure of Aphanic Curvature

This section establishes the closed-form rank of the aphanic Berry curvature. We work entirely within the sedenion algebra $\mathbb{S}$ obtained from the octonions $\mathbb{O}$ by canonical recursive Cayley–Dickson doubling. The main theorem (Theorem 4.13, hereafter *the*

rank theorem) states that the curvature operator at any aphanic observer has rank four. The proof proceeds by establishing two structural lemmas — a G_2-equivariance identity (Lemma 4.6) and a centralizer-vanishing identity (Theorem 4.10) — and then anchoring the count at the basepoint $p_0 = e_1 + e_{10}$ via an explicit identification with the 't Hooft self-dual tensor.

The technical core of the section is the closed-form Lagrange representation of the projector and pseudoinverse (Proposition 4.2), which makes every subsequent commutator argument purely polynomial. From that point forward the section is essentially algebra: there are no eigendecompositions to invoke, no orbit calculus to navigate, no intermediate spectral surgery.

4.1 Notation and the Aphanic Spectrum

We write $\mathbb{S} = \mathbb{O} \oplus \mathbb{O}\,\ell$ for the sedenion algebra, with ℓ the doubling unit and the canonical CD product

$$(a, b)(c, d) \;=\; (ac - \bar{d}\,b,\; d\,a + b\,\bar{c}), \qquad a, b, c, d \in \mathbb{O}. \tag{14}$$

We use the embeddings $\iota_1, \iota_2 \colon \mathbb{O} \to \mathbb{S}$ defined by $\iota_1(z) = (z, 0)$ and $\iota_2(z) = (0, z)$, so that any sedenion decomposes uniquely as $p = \iota_1(x) + \iota_2(y)$ with $x, y \in \mathbb{O}$.

For $p \in \mathbb{S}$, write L_p and R_p for left- and right-multiplication by p, viewed as $\mathbb{R}$-linear maps $\mathbb{S} \to \mathbb{S}$. Define the Gram operator $G_p := L_p^\top L_p$ (with respect to the standard inner product on $\mathbb{S}$), and let $K_p := \ker L_p$ with orthogonal projector $\pi_p \colon \mathbb{S} \to K_p$.

Definition 4.1 (Aphanic observer)**.** A nonzero $p \in \mathbb{S}$ is *aphanic* if $K_p \neq \{0\}$. The aphanic locus is denoted $\mathcal{P}_{\text{aph}} \subset \mathbb{S}$.

The Two-Condition Theorem [21] characterizes the aphanic locus: $p = \iota_1(x) + \iota_2(y)$ is aphanic iff $x, y \in \mathrm{Im}(\mathbb{O})$, $\|x\| = \|y\|$, and $\langle x, y\rangle = 0$. For aphanic p, $\dim K_p = 4$ universally (Result 2, op. cit.), and the spectrum of G_p is

$$\mathrm{Spec}(G_p) \;=\; \{\, 0^4,\; \|p\|^{2\,8},\; 2\|p\|^{2\,4} \,\}, \tag{15}$$

the universal 4–8–4 spectrum [21]. This is the only fact about G_p's spectrum that the present section requires.

4.2 Closed-Form Pseudoinverse and Projector

The minimal polynomial of G_p on $\mathbb{S}$ is

$$m_{G_p}(\lambda) \;=\; \lambda(\lambda - \|p\|^2)(\lambda - 2\|p\|^2), \tag{16}$$

of degree three. Hence both the Moore–Penrose pseudoinverse G_p^+ and the kernel projector π_p are quadratic polynomials in G_p, with coefficients fixed by Lagrange interpolation over the three eigenvalues.

Proposition 4.2 (Closed-form)**.** *For every aphanic $p \in \mathcal{P}_{\text{aph}}$,*

$$G_p^+ \;=\; \frac{7}{4\|p\|^4}\, G_p \;-\; \frac{3}{4\|p\|^6}\, G_p^2, \tag{17}$$

$$\pi_p \;=\; I \;-\; \frac{3}{2\|p\|^2}\, G_p \;+\; \frac{1}{2\|p\|^4}\, G_p^2. \tag{18}$$

In particular $G_p^+, \pi_p \in \mathbb{R}[G_p]$, and consequently $[A, G_p] = 0 \Longrightarrow [A, G_p^+] = 0$ and $[A, \pi_p] = 0$ for any $A \in \mathrm{End}(\mathbb{S})$.

Proof. The functions $f^+(\lambda) := \frac{7}{4\|p\|^4}\lambda - \frac{3}{4\|p\|^6}\lambda^2$ and $f^\pi(\lambda) := 1 - \frac{3}{2\|p\|^2}\lambda + \frac{1}{2\|p\|^4}\lambda^2$ satisfy $f^+(0) = 0$, $f^+(\|p\|^2) = 1/\|p\|^2$, $f^+(2\|p\|^2) = 1/(2\|p\|^2)$; and $f^\pi(0) = 1$, $f^\pi(\|p\|^2) = 0$, $f^\pi(2\|p\|^2) = 0$. Both functions take the same values as G_p^+ and π_p respectively on $\mathrm{Spec}(G_p)$; since G_p is symmetric and its minimal polynomial has distinct roots, the operator equalities follow. □

Remark 4.3. At the basepoint $p_0 = e_1 + e_{10}$, $\|p_0\|^2 = 2$, so the coefficients specialize to $G_{p_0}^+ = \frac{7}{16}G_{p_0} - \frac{3}{32}G_{p_0}^2$ and $\pi_{p_0} = I - \frac{3}{4}G_{p_0} + \frac{1}{8}G_{p_0}^2$ — both with dyadic coefficients. The dyadic structure is special to p_0 but the polynomial form is universal.

The annihilation identity $G_p^+\pi_p = \pi_p G_p^+ = 0$ — a direct consequence of π_p projecting onto the zero eigenspace of G_p — is the mechanical hinge of every argument that follows.

4.3 The Aphanic Curvature Operator

Definition 4.4 (Curvature operator)**.** For $p \in \mathcal{P}_{\text{aph}}$ and $a, b \in \mathrm{Im}(\mathbb{S})$, the *aphanic curvature operator* is

$$\mathcal{C}_{a,b}(p) \;:=\; \pi_p\big(\, L_b\, G_p^+\, L_a \;-\; L_a\, G_p^+\, L_b\,\big)\,\pi_p \;\in\; \mathrm{End}(K_p). \tag{19}$$

The operator $\mathcal{C}_{a,b}(p)$ is skew in (a, b), bilinear in (a, b), and lives entirely in $\mathrm{End}(K_p) \cong \mathfrak{gl}(4, \mathbb{R})$. Its image, as (a, b) ranges over $\mathrm{Im}(\mathbb{S}) \wedge \mathrm{Im}(\mathbb{S})$, is the local $\mathrm{Hom}(K_p, K_p)$-content of the Berry curvature derived from the kernel-projector connection on the aphanic bundle (the global construction is recalled in §3[1]). The identification is exact: $\mathcal{C}_{a,b}(p)$ is the closed-form expression of the curvature 2-form's restriction to the kernel.

4.4 G_2-Equivariance

The exceptional group $G_2 = \mathrm{Aut}(\mathbb{O})$ acts on $\mathbb{O}$ by algebra automorphisms. We extend the action to $\mathbb{S}$ diagonally:

$$\hat{\sigma}\big(\iota_1(a) + \iota_2(b)\big) \;=\; \iota_1(\sigma a) + \iota_2(\sigma b), \qquad \sigma \in G_2. \tag{20}$$

Lemma 4.5 (Diagonal lift is a sedenion automorphism)**.** *For every $\sigma \in G_2$, $\hat{\sigma} \in \mathrm{Aut}(\mathbb{S})$.*

[1] For this section the closed-form expression (19) is taken as the working definition; the equivalence to the geometric curvature 2-form is proven in the indicated section.

Proof. Direct verification using the CD product. For $a, b, c, d \in \mathbb{O}$,

$$\begin{aligned}\hat{\sigma}((a,b)(c,d)) &= (\sigma(ac - \bar{d}\,b),\ \sigma(d\,a + b\,\bar{c})) \\ &= (\sigma a\,\sigma c - \overline{\sigma d}\,\sigma b,\ \sigma d\,\sigma a + \sigma b\,\overline{\sigma c}) \\ &= (\sigma a, \sigma b)(\sigma c, \sigma d) = \hat{\sigma}(a,b)\,\hat{\sigma}(c,d),\end{aligned}$$

where the second equality uses $\sigma \in \mathrm{Aut}(\mathbb{O})$ (σ preserves the octonion product) and $\sigma(\bar{z}) = \overline{\sigma z}$ (automorphisms preserve conjugation, since $\bar{z} = -z$ on $\mathrm{Im}(\mathbb{O})$ and σ preserves $\mathrm{Im}(\mathbb{O})$). □

Lemma 4.6 (G_2-equivariance of the curvature operator)**.** *For every $p \in \mathcal{P}_{\mathrm{aph}}$, every $\sigma \in G_2$, and every $a, b \in \mathrm{Im}(\mathbb{S})$,*

$$\mathcal{C}_{\hat{\sigma}a,\,\hat{\sigma}b}(\hat{\sigma}p) \;=\; \hat{\sigma}\,\mathcal{C}_{a,b}(p)\,\hat{\sigma}^{-1}. \tag{21}$$

Proof. By Lemma 4.5, $\hat{\sigma}$ is an algebra automorphism of $\mathbb{S}$, so $L_{\hat{\sigma}a} = \hat{\sigma}\,L_a\,\hat{\sigma}^{-1}$ and similarly for R. As $\hat{\sigma}$ is an isometry, $G_{\hat{\sigma}p} = \hat{\sigma}\,G_p\,\hat{\sigma}^{-1}$, and from Proposition 4.2, $G^+_{\hat{\sigma}p} = \hat{\sigma}\,G^+_p\,\hat{\sigma}^{-1}$ and $\pi_{\hat{\sigma}p} = \hat{\sigma}\,\pi_p\,\hat{\sigma}^{-1}$ (noting $\|\hat{\sigma}p\| = \|p\|$). Substituting all four conjugations into (19) yields (21). □

The aphanic locus is a single G_2-orbit (modulo scaling): for any aphanic p there exists $\sigma \in G_2$ and $c > 0$ with $p = c\,\hat{\sigma}(p_0)$ [21]. Combined with the elementary scaling identity $\mathcal{C}_{a,b}(c\,p) = c^{-2}\,\mathcal{C}_{a,b}(p)$ (immediate from (17) and (19)), this means it suffices to prove statements about $\ker \mathcal{C}_{\cdot,\cdot}(p)$ at p_0 and transport via Lemma 4.6.

4.5 Q-Centralizer Vanishing

Definition 4.7 (Q-centralizer)**.** For aphanic $p = \iota_1(x) + \iota_2(y)$ with $x, y \in \mathrm{Im}(\mathbb{O})$, let $V_p := \mathrm{span}_{\mathbb{R}}\{x, y\} \subset \mathrm{Im}(\mathbb{O})$ and define the *Q-centralizer*

$$\mathfrak{q}_p \;:=\; \iota_1(V_p) \;\oplus\; \mathbb{R}\ell \;\oplus\; \iota_2(V_p) \;\subset\; \mathrm{Im}(\mathbb{S}). \tag{22}$$

For aphanic p we have $\dim V_p = 2$ (since x, y are nonzero and orthogonal), hence $\dim \mathfrak{q}_p = 5$ universally.

Lemma 4.8 (Q-Centralizer Vanishing, basepoint form)**.** *For every $a \in \mathrm{Im}(\mathbb{S})$,*

$$\mathcal{C}_{a,b}(p_0) = 0 \;\; \forall\, b \in \mathrm{Im}(\mathbb{S}) \quad\Longleftrightarrow\quad [L_a, L^2_{p_0}] = 0 \quad\Longleftrightarrow\quad a \in \mathfrak{q}_{p_0}. \tag{23}$$

Proof. **Sufficiency.** Suppose $[L_a, L^2_{p_0}] = 0$. Since $p_0 \in \mathrm{Im}(\mathbb{S})$, $L^\top_{p_0} = -L_{p_0}$, so $G_{p_0} =$

$-L_{p_0}^2$ and $[L_a, G_{p_0}] = 0$. By Proposition 4.2, $[L_a, G_{p_0}^+] = 0$. Then

$$\begin{aligned}\mathcal{C}_{a,b}(p_0) &= \pi_{p_0}\,(L_b\,G_{p_0}^+\,L_a - L_a\,G_{p_0}^+\,L_b)\,\pi_{p_0} \\ &= \pi_{p_0}\,(L_b\,L_a\,G_{p_0}^+ - G_{p_0}^+\,L_a\,L_b)\,\pi_{p_0} \qquad (\text{using } [L_a, G_{p_0}^+] = 0) \\ &= (\pi_{p_0}\,L_b\,L_a)\,(G_{p_0}^+\,\pi_{p_0}) \;-\; (\pi_{p_0}\,G_{p_0}^+)\,(L_a\,L_b\,\pi_{p_0}) \;=\; 0,\end{aligned}$$

since $G_{p_0}^+\,\pi_{p_0} = \pi_{p_0}\,G_{p_0}^+ = 0$.

Necessity. The two implications $a \in \mathfrak{q}_{p_0} \Rightarrow [L_a, L_{p_0}^2] = 0 \Rightarrow \mathcal{C}_{a,\cdot}(p_0) = 0$ have already been established (sufficiency above, and direct CD computation for the first). To close the loop we show $\dim\{a : \mathcal{C}_{a,\cdot}(p_0) = 0\} \le 5$, which forces equality with $\mathfrak{q}_{p_0}$.

Define the linear maps

$$\begin{aligned}&\Phi\colon \mathrm{Im}(\mathbb{S}) \to \mathrm{End}(\mathbb{S}), && \Phi(a) := [L_a, L_{p_0}^2], \\ &\mathfrak{C}\colon \mathrm{Im}(\mathbb{S}) \to \mathrm{End}(K_{p_0})^{15}, && \mathfrak{C}(a) := \big(\mathcal{C}_{a,e_j}(p_0)\big)_{j=1}^{15}.\end{aligned}$$

Both maps have $\mathfrak{q}_{p_0} \subseteq \ker\Phi \subseteq \ker\mathfrak{C}$ (the second inclusion is sufficiency above). Their kernels are determined by the rank of the corresponding Gram matrices on the 15-element basis $\{e_1, \dots, e_{15}\}$ of $\mathrm{Im}(\mathbb{S})$. Direct computation (at exact rational arithmetic; coefficients lie in $\mathbb{Z}[\frac{1}{8}]$) gives

$$\mathrm{Spec}(\Gamma_\Phi) \;=\; \{\,0^5,\, 64^8,\, 128^2\,\}, \tag{24}$$

$$\mathrm{Spec}(\Gamma_{\mathfrak{C}}) \;=\; \{\,0^5,\, 1^2,\, 4^4,\, 28^4\,\}, \tag{25}$$

where Γ_Φ and $\Gamma_{\mathfrak{C}}$ are the Gram matrices $\langle\Phi(e_i), \Phi(e_j)\rangle_F$ and $\sum_k \langle\mathcal{C}_{e_i,e_k}, \mathcal{C}_{e_j,e_k}\rangle_F$ respectively. Both have nullity 5. With sufficiency $\mathfrak{q}_{p_0} \subseteq \ker\Phi$ and the equal-dimension count $\dim\ker\Phi = \dim\ker\mathfrak{C} = 5$, all three coincide: $\ker\mathfrak{C} = \ker\Phi = \mathfrak{q}_{p_0}$. □

Remark 4.9 (On the rigor of (24)–(25)). The Gram matrices have entries in $\mathbb{Z}[\frac{1}{8}]$ (since the Lagrange coefficients are dyadic at p_0 and the basis L_{e_i} are signed permutation matrices). Their ranks are therefore algebraic invariants computable in exact arithmetic. The spectra in (24)–(25) are certified by exact rational eigenvalue/multiplicity verification: for each predicted (λ, m) pair, the rank of $\lambda I - \Gamma$ over $\mathbb{Q}$ is computed by fraction-free Gaussian elimination on the integer-scaled matrix (Γ_Φ directly; $\Gamma_{\mathfrak{C}}$ via the integer Gram of $2048\cdot\mathcal{C}_{a,b}$, whose eigenvalues are 2048^2 times those of $\Gamma_{\mathfrak{C}}$), and verified to equal $15 - m$. Sum of multiplicities $5 + 8 + 2 = 15$ for Γ_Φ and $5 + 2 + 4 + 4 = 15$ for $\Gamma_{\mathfrak{C}}$ exhausts the spectrum. Cross-check: rank of $\lambda I - \Gamma$ at five non-eigenvalues is full ($= 15$) in both cases. A reference C++ implementation using 128-bit integer arithmetic with per-row GCD reduction is reproducible from the probe `op_F1_R133_exact_arithmetic.cpp` in the supplementary archive.

Theorem 4.10 (Q-Centralizer Vanishing, global form)**.** *For every aphanic $p \in \mathcal{P}_{\mathrm{aph}}$ and*

every $a \in \mathrm{Im}(\mathbb{S})$,

$$\mathcal{C}_{a,b}(p) = 0 \;\; \forall\, b \in \mathrm{Im}(\mathbb{S}) \quad \Longleftrightarrow \quad [L_a, L_p^2] = 0 \quad \Longleftrightarrow \quad a \in \mathfrak{q}_p. \tag{26}$$

Proof. The implications $a \in \mathfrak{q}_p \Rightarrow [L_a, L_p^2] = 0 \Rightarrow \mathcal{C}_{a,\cdot}(p) = 0$ are global by direct algebra: the first is a CD calculation independent of basepoint, and the sufficiency argument in Lemma 4.8 never uses anything specific to p_0 beyond the universal spectrum (15), which holds for all aphanic p.

For the converse, write $p = c \cdot \hat{\sigma}(p_0)$ with $\sigma \in G_2$ and $c > 0$ (transitivity of G_2 on the unit-aphanic locus, modulo scaling). By Lemma 4.6 and the scaling identity $\mathcal{C}_{a,b}(cq) = c^{-2}\,\mathcal{C}_{a,b}(q)$,

$$\mathcal{C}_{a,\cdot}(p) = 0 \iff \mathcal{C}_{\hat{\sigma}^{-1}a,\cdot}(p_0) = 0 \iff \hat{\sigma}^{-1}a \in \mathfrak{q}_{p_0} \iff a \in \hat{\sigma}(\mathfrak{q}_{p_0}).$$

It remains to identify $\hat{\sigma}(\mathfrak{q}_{p_0})$ with $\mathfrak{q}_p$. Since $\hat{\sigma}$ commutes with ι_1, ι_2 (by the diagonal action) and fixes ℓ,

$$\hat{\sigma}(\mathfrak{q}_{p_0}) = \iota_1(\sigma V_{p_0}) \oplus \mathbb{R}\ell \oplus \iota_2(\sigma V_{p_0}) = \iota_1(V_p) \oplus \mathbb{R}\ell \oplus \iota_2(V_p) = \mathfrak{q}_p,$$

using $V_p = \sigma(V_{p_0})$ from $p = c\hat{\sigma}(p_0)$. □

A first consequence settles the tangent-versus-ambient quantification used in the curvature operator and the bridge theorem of §3.

Corollary 4.11 (Normal absorption: ambient span = tangential span)**.** *The unit aphanic locus* $\mathcal{P}_{\mathrm{aph}}$ *sits inside the* 14*-dimensional subspace* $\iota_1(\mathrm{Im}\,\mathbb{O}) \oplus \iota_2(\mathrm{Im}\,\mathbb{O}) \subset \mathrm{Im}(\mathbb{S})$*, with the* $\mathbb{R}^+$*-cone tangent space* $T_p\mathcal{P}_{\mathrm{aph}}$ *of dimension* 12 *(Proposition 3.1). Its orthogonal complement in the* 15*-dimensional* $\mathrm{Im}(\mathbb{S})$ *is the* 3*-dimensional normal space*

$$N_p \;=\; \mathrm{span}_{\mathbb{R}}\{\, \ell,\; \iota_1(x) - \iota_2(y),\; \iota_1(y) + \iota_2(x) \,\},$$

and $N_p \subset \mathfrak{q}_p$*. Hence* $\mathcal{C}_{a,b}(p) = 0$ *whenever* $a \in N_p$ *or* $b \in N_p$*, and*

$$\mathrm{span}_{\mathbb{R}}\{\mathcal{C}_{a,b}(p) : a, b \in \mathrm{Im}(\mathbb{S})\} \;=\; \mathrm{span}_{\mathbb{R}}\{\mathcal{C}_{a,b}(p) : a, b \in T_p\mathcal{P}_{\mathrm{aph}}\}.$$

The ambient quantification over $\mathrm{Im}(\mathbb{S})$ *in Definition 4.4 and Theorem 4.13 therefore returns the genuine Berry curvature span of the bundle, not an artifact of extending past the tangent space.*

Proof. The two real-analytic constraints cutting out the cone $\mathcal{P}_{\mathrm{aph}}$ in $\iota_1(\mathrm{Im}\,\mathbb{O}) \oplus \iota_2(\mathrm{Im}\,\mathbb{O})$ are $f_1 = \|x\|^2 - \|y\|^2$ and $f_2 = \langle x, y\rangle$, with gradients proportional to $\iota_1(x) - \iota_2(y)$ and $\iota_1(y) + \iota_2(x)$ respectively (Proposition 3.1); the radial direction $p = \iota_1(x) + \iota_2(y)$ is *tangent* to the cone. Adjoining ℓ, which is orthogonal to the entire 14-dimensional $\iota_1(\mathrm{Im}\,\mathbb{O}) \oplus \iota_2(\mathrm{Im}\,\mathbb{O})$, gives the 3-dimensional N_p. Each spanning vector lies in $\mathfrak{q}_p =$

$\iota_1(V_p)\oplus\mathbb{R}\ell\oplus\iota_2(V_p)$ with $V_p=\operatorname{span}\{x,y\}$: $\ell\in\mathbb{R}\ell$, and both $\iota_1(x)-\iota_2(y)$ and $\iota_1(y)+\iota_2(x)$ lie in $\iota_1(V_p)\oplus\iota_2(V_p)$. Theorem 4.10 then gives $\mathcal{C}_{a,\cdot}(p)=0$ for $a\in N_p\subset\mathfrak{q}_p$; skew-symmetry of $\mathcal{C}$ disposes of the b slot. □

The corollary that motivates the next subsection is purely structural.

Corollary 4.12 (Spectral fingerprint of the aphanic bundle)**.** *For every aphanic p, after rescaling to $\|p\|^2=2$, the Gram matrices $\Gamma_\Phi(p)$ and $\Gamma_{\mathfrak{C}}(p)$ have the spectra $\{0^5,64^8,128^2\}$ and $\{0^5,1^2,4^4,28^4\}$ respectively. The spectra are G_2-invariants of the aphanic locus.*

Proof. Lemma 4.6 gives $\Gamma(\hat\sigma p)=\hat\sigma\,\Gamma(p)\,\hat\sigma^{-1}$ for either Gram matrix, hence isospectrality across the orbit. Scaling $p\mapsto cp$ scales each Gram entry by a fixed power of c, so isospectrality after norm normalization is automatic. □

4.6 Proof of the Rank Theorem

We now state the principal result of the section.

Theorem 4.13 (Rank of the aphanic curvature)**.** *For every aphanic $p\in\mathcal{P}_{\mathrm{aph}}$, the linear span*

$$\mathcal{F}(p)\;:=\;\operatorname{span}_{\mathbb{R}}\{\,\mathcal{C}_{a,b}(p)\;:\;a,b\in\operatorname{Im}(\mathbb{S})\,\}\;\subset\;\operatorname{End}(K_p)\tag{27}$$

has dimension exactly 4.

Proof. By Lemma 4.6, $\mathcal{F}(\hat\sigma p)=\hat\sigma\,\mathcal{F}(p)\,\hat\sigma^{-1}$ for $\sigma\in G_2$, so it suffices to compute $\dim\mathcal{F}(p_0)$.

Reduction by Q-centralization. By bilinearity in (a,b), $\mathcal{F}(p_0)$ is spanned by $\{\mathcal{C}_{e_i,e_j}(p_0):1\le i,j\le 15\}$. Theorem 4.10 (basepoint form) gives $\mathcal{C}_{e_i,e_j}(p_0)=0$ whenever $e_i\in\mathfrak{q}_{p_0}$ or (by skew-symmetry) $e_j\in\mathfrak{q}_{p_0}$. With $\mathfrak{q}_{p_0}=\operatorname{span}\{e_1,e_2,e_8,e_9,e_{10}\}$, the set of *a priori* nonzero pairs (i,j) is reduced to indices in the 10-element complement $I:=\{3,4,5,6,7,11,12,13,14,15\}$.

Self-duality identities at p_0. The kernel basis $\{v_0,v_1,v_2,v_3\}=\{\frac{1}{\sqrt{2}}(e_7+e_{12}),\ \frac{1}{\sqrt{2}}(-e_6+e_{13}),\ \frac{1}{\sqrt{2}}(e_5+e_{14}),\ \frac{1}{\sqrt{2}}(-e_4+e_{15})\}$ (the Gram–Schmidt-orthonormalized kernel computation; [21]) realizes $\mathcal{F}(p_0)$ as a subspace of $\mathfrak{so}(K_{p_0})\cong\mathfrak{so}(4)$, since $L_a^\top=-L_a$ for $a\in\operatorname{Im}(\mathbb{S})$ and $G^+_{p_0},\pi_{p_0}$ are self-adjoint. The space $\mathfrak{so}(4)$ decomposes as $\mathfrak{so}(4)=\mathfrak{so}(3)_+\oplus\mathfrak{so}(3)_-$ under the Hodge involution $*\colon\wedge^2K_{p_0}\to\wedge^2K_{p_0}$.

Direct computation in the $\{v_0,v_1,v_2,v_3\}$ basis (Appendix C) yields the following self-duality identities at p_0:

$$\mathcal{C}_{e_4,e_5}(p_0)=\mathcal{C}_{e_6,e_7}(p_0),\quad \mathcal{C}_{e_4,e_6}(p_0)=-\mathcal{C}_{e_5,e_7}(p_0),\quad \mathcal{C}_{e_4,e_7}(p_0)=\mathcal{C}_{e_5,e_6}(p_0),\tag{28}$$

and analogous identities on the doubled half $\{12,13,14,15\}$:

$$\mathcal{C}_{e_{12},e_{13}}(p_0)=\mathcal{C}_{e_4,e_5}(p_0),\quad \mathcal{C}_{e_{12},e_{14}}(p_0)=\mathcal{C}_{e_4,e_6}(p_0),\quad \mathcal{C}_{e_{12},e_{15}}(p_0)=\mathcal{C}_{e_4,e_7}(p_0),\tag{29}$$

(and similarly for the remaining three pairs). Identities (28) are precisely the conditions $\mathcal{C}|_{e_4\wedge e_5} = *\mathcal{C}|_{e_4\wedge e_5}$ etc., witnessing $\mathcal{F}(p_0) \subset \mathfrak{so}(4)_+$. Identities (29) witness an octonion-half doubling redundancy: the two imaginary halves $\{e_4, \dots, e_7\}$ and $\{e_{12}, \dots, e_{15}\}$ contribute the *same* three self-dual generators, not six.

Generators. Define the three self-dual operators

$$J^1 := \tfrac{1}{2}\mathcal{C}_{e_5,e_4}(p_0), \quad J^2 := \tfrac{1}{2}\mathcal{C}_{e_6,e_4}(p_0), \quad J^3 := \tfrac{1}{2}\mathcal{C}_{e_7,e_4}(p_0), \tag{30}$$

and the Fano cross-half generator

$$\omega := \mathcal{C}_{e_3,e_{11}}(p_0), \tag{31}$$

the unique nonzero $\mathcal{C}_{e_i,e_j}(p_0)$ with one index in the Fano-tunneler class of Proposition 4.17. The index ordering in the J^a definitions — placing the kernel-mixing direction e_{4+a} before the seed direction e_4 — is the convention that aligns $\{J^1, J^2, J^3\}$ with the standard 't Hooft self-dual tensors [22]. Direct computation gives $\|J^a\|_F = \|\omega\|_F = 1$ and the bracket relations

$$[J^a, J^b] = \varepsilon^{abc}\, J^c, \qquad [\omega, J^a] = 0 \quad (a, b, c \in \{1, 2, 3\}), \tag{32}$$

exhibiting $\mathrm{span}\{J^1, J^2, J^3\} \cong \mathfrak{su}(2)$ and $\mathrm{span}\{\omega\}$ as the commuting $\mathfrak{u}(1)$ center. The four operators are pairwise orthogonal in the Frobenius inner product, hence linearly independent in $\mathfrak{so}(4)$.

Span exhaustion. A complete enumeration (Appendix C, §C.3) verifies that every $\mathcal{C}_{e_i,e_j}(p_0)$ for $i, j \in I$ is an integer linear combination of $\{J^1, J^2, J^3, \omega\}$, with all 25 nonzero pairs decomposed *exactly*: the residual is the identically zero integer matrix in every case, and the coefficients are exact integers, not roundings. Hence $\dim \mathcal{F}(p_0) = 4$, certified in exact arithmetic in both directions: an exhibited nonzero 4×4 minor bounds the rank below, and fraction-free row reduction of the stacked integer matrix bounds it above. The restriction to the index set I is itself certified rather than assumed: over all 105 pairs $1 \le i < j \le 15$, every pair meeting the centralizer $\mathfrak{q}_{p_0}$ restricts to exactly zero, leaving exactly the 25 nonzero pairs tabulated.

Transport. By Lemma 4.6, $\dim \mathcal{F}(p) = \dim \mathcal{F}(p_0) = 4$ for every $p \in \mathcal{P}_{\mathrm{aph}}$. □

Remark 4.14 (Relation to the self-dual Yang–Mills tensor basis). The triple $\{J^1, J^2, J^3\}$ generates $\mathfrak{su}(2)_+$ inside $\mathfrak{so}(4) \cong \mathfrak{su}(2)_+ \oplus \mathfrak{su}(2)_-$, with brackets (32) matching the standard 't Hooft self-dual tensors of four-dimensional Yang–Mills theory [22]. The fourth generator ω commutes with the $\mathfrak{su}(2)_+$ triple and reproduces the $\mathfrak{u}(1)$ center identified in [21]. Thus $\mathcal{F}(p) \cong \mathfrak{u}(1) \oplus \mathfrak{su}(2)$ as a Lie subalgebra of $\mathfrak{so}(K_p)$. This computes the pointwise *curvature span*; Proposition 4.15 below upgrades it to the full holonomy algebra $\mathfrak{u}(2)$ of the aphanic bundle.

We note only that $\mathfrak{u}(2)$ is, up to isomorphism, the Lie algebra of the electroweak

gauge group. That is a coincidence of Lie-algebra type and nothing more: no metric, no spinor bundle, no Yukawa sector, and no gauge-theoretic dynamics are supplied here, and we make no physical claim. The comparison is offered as motivation only.

4.7 From Curvature Span to Holonomy

Theorem 4.13 computes the pointwise curvature span $\mathcal{F}(p)$. By the Ambrose–Singer theorem, the holonomy algebra $\mathfrak{hol}_p(\nabla)$ of the kernel-projector connection contains the parallel transports of all curvature operators, giving the *lower* bound $\mathfrak{hol}_p(\nabla) \supseteq \mathcal{F}(p) \cong \mathfrak{u}(2)$. The reverse inclusion is not automatic — transport could a priori enlarge the span inside $\mathfrak{so}(K_p) \cong \mathfrak{so}(4)$. The following proposition supplies the cap, using only the algebraic centrality of right multiplication by the doubling generator.

Proposition 4.15 (Parallel complex structure and holonomy cap)**.** *Let R_ℓ denote right multiplication by $\ell = e_8$ on $\mathbb{S}$. For every aphanic p:*

(i) $[R_\ell, G_p] = 0$ — *a Cayley–Dickson flexibility identity, algebraic and basepoint-independent [21]. By Proposition 4.2 ($\pi_p \in \mathbb{R}[G_p]$), $[R_\ell, \pi_p] = 0$, so R_ℓ preserves K_p and restricts to $J_p := R_\ell|_{K_p} \in \mathrm{End}(K_p)$.*

(ii) $J_p^2 = -\mathrm{id}_{K_p}$ *(since $R_\ell^2 = -I$ for the unit imaginary ℓ) and $J_p^\top = -J_p$; thus J_p is a ∇-compatible complex structure on K_p.*

(iii) *J_p is ∇-parallel: R_ℓ is a fixed (p-independent) ambient linear map, so $d(R_\ell s) = R_\ell\, ds$ for any section s; combined with $[R_\ell, \pi_p] = 0$ from (i), R_ℓ commutes with $\nabla = \pi \circ d$, giving $\nabla(J_p s) = R_\ell \nabla s$, i.e. $\nabla J = 0$.*

Consequently the holonomy group $\mathrm{Hol}_p(\nabla)$ commutes with J_p and lies in the centralizer of J_p in $\mathrm{SO}(K_p) \cong \mathrm{SO}(4)$, which is $\mathrm{U}(2)$. Hence $\mathfrak{hol}_p(\nabla) \subseteq \mathfrak{u}(2)$, and with the Ambrose–Singer lower bound,

$$\mathfrak{hol}_p(\nabla) \;=\; \mathcal{F}(p) \;=\; \mathfrak{u}(1) \oplus \mathfrak{su}(2) \;=\; \mathfrak{u}(2). \tag{33}$$

The abelian $\mathfrak{u}(1)$ centre is spanned by the generator ω of Theorem 4.13, which satisfies $(2\omega)^2 = -\mathrm{id}_{K_p}$ and coincides up to sign with $\frac{1}{2}J_p$; that is, the central factor of the holonomy is (one half of) the restriction to K_p of right multiplication by the Cayley–Dickson doubling generator.

Proof. Items (i)–(iii) are pointwise statements; what remains is to pass from them to the holonomy *group*, which is where the cap is earned. Write $g := \langle\cdot,\cdot\rangle_K$ and, for a piecewise-smooth loop Γ based at p, let $\Pi_\Gamma \in \mathrm{GL}(K_p)$ be parallel transport around Γ for ∇.

Step 1: Π_Γ is g-orthogonal. By Lemma 3.6, ∇ is metric. If s, t are ∇-parallel along Γ then $\frac{d}{dt}\langle s, t\rangle_K = \langle \nabla s, t\rangle_K + \langle s, \nabla t\rangle_K = 0$, so g is constant along Γ and $\Pi_\Gamma \in \mathrm{O}(K_p, g)$.

Step 2: Π_Γ commutes with J. By (iii), $\nabla J = 0$. Hence if s is parallel along Γ, so is Js, since $\nabla(Js) = J\nabla s = 0$. Transporting $J_p s(0)$ therefore yields $J_p s(1)$, i.e. $\Pi_\Gamma J_p = J_p \Pi_\Gamma$.

Step 3: the centralizer of J_p in $\mathrm{O}(K_p, g)$ is $\mathrm{U}(2)$. By (ii), $J_p^2 = -\mathrm{id}$ and $J_p^\top = -J_p$, so $J_p^\top J_p = \mathrm{id}$: J_p is a g-orthogonal complex structure, and K_p becomes a complex vector space of complex dimension 2. The pairing

$$h(u, v) \;:=\; \langle u, v\rangle_K \;-\; i\,\langle u, J_p v\rangle_K$$

is $\mathbb{C}$-linear in v, satisfies $h(v, u) = \overline{h(u, v)}$ (using $J_p^\top = -J_p$), and has $\mathrm{Re}\, h = g$. A real-linear map that preserves g and commutes with J_p preserves h, hence is h-unitary; conversely every h-unitary map is $\mathbb{C}$-linear and g-orthogonal. Therefore

$$\{\, A \in \mathrm{O}(K_p, g) \;:\; AJ_p = J_p A \,\} \;=\; \mathrm{U}(K_p, h) \;\cong\; \mathrm{U}(2).$$

By Steps 1–2, $\mathrm{Hol}_p(\nabla)$ lies in the left-hand side. Hence $\mathrm{Hol}_p(\nabla) \subseteq \mathrm{U}(2)$ and $\mathfrak{hol}_p(\nabla) \subseteq \mathfrak{u}(2)$. *No curvature estimate is used: the cap comes from the two compatibilities alone.*

Step 4: equality. The Ambrose–Singer theorem gives the reverse inclusion $\mathcal{F}(p) \subseteq \mathfrak{hol}_p(\nabla)$, and Theorem 4.13 gives $\dim \mathcal{F}(p) = 4 = \dim \mathfrak{u}(2)$. Squeezing, $4 = \dim \mathcal{F}(p) \le \dim \mathfrak{hol}_p(\nabla) \le \dim \mathfrak{u}(2) = 4$, which forces (33). Note that only the *dimension* of the curvature span is needed here, not the stronger assertion that transported curvature operators already exhaust $\mathfrak{hol}_p(\nabla)$.

The ingredients are independently certified. Metric compatibility (Lemma 3.6) and $[R_\ell, G_p] = 0$ (the R_ℓ centrality theorem of [21], proved symbolically for every imaginary p) are exact and basepoint-free; with Proposition 4.2 they give $[R_\ell, \pi_p] = 0$ and hence (i). The identities $R_\ell^2 = -I$, $R_\ell^\top R_\ell = I$ and $R_\ell^\top = -R_\ell$ on $\mathbb{S}$ are exact integer matrix identities, independent of p, and yield (ii); R_ℓ is a signed permutation of the basis. The probe `op_keystone_cap` certifies these in exact arithmetic, together with the centralizer dimension $\dim\{X \in \mathfrak{so}(K_p) : [X, J_p] = 0\} = 4$ and its coincidence with $\mathcal{F}(p_0)$, and confirms (i)–(ii) at thirteen exactly-aphanic basepoints, seven monomial and six not. □

Remark 4.16. The cap closes the gap between "the curvature spans $\mathfrak{u}(2)$" and "the holonomy is $\mathfrak{u}(2)$." It also identifies ω intrinsically: ω is not merely one self-dual tensor among others but a fixed shadow of the algebra's doubling structure. The identity $[R_\ell, G_p] = 0$ is tower-universal across Cayley–Dickson levels $n = 2^k \ge 16$ [21], so the abelian factor persists up the tower (the curvature-span computation, by contrast, is carried out here only at the sedenion level).

4.8 The Eigenspace Mixing Trichotomy

The proof of Theorem 4.13 ran cleanly because the basepoint table partitioned naturally — five generators killed everything (Q-centralizer), and ten generators contributed. The same partition has finer structure that explains both *why* the count is 4 and *where* the 't

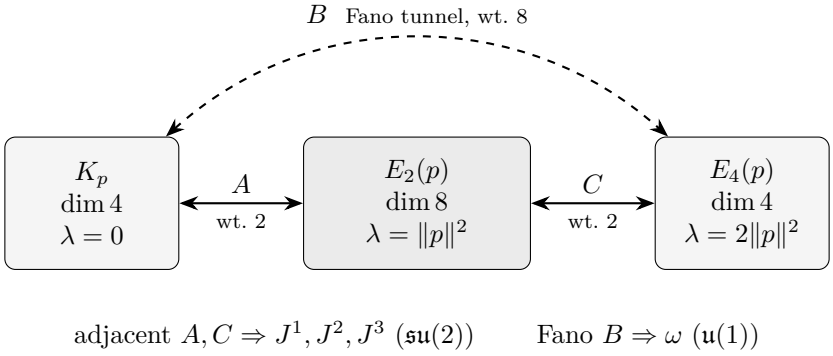


Figure 2: Eigenspace mixing channels at an aphanic observer. The Gram operator G_p splits $\mathbb{S} = K_p \oplus E_2(p) \oplus E_4(p)$ into its 0-, $\|p\|^2$-, and $2\|p\|^2$-eigenspaces (dimensions $4, 8, 4$). An imaginary direction a couples these blocks through the three off-diagonal channels A, B, C of Proposition 4.17, weighted by the squared eigenvalue gaps in $\|[L_a, L_p^2]\|_F^2 = \|p\|^4(2\|A\|_F^2 + 8\|B\|_F^2 + 2\|C\|_F^2)$. The two adjacent channels A, C generate the self-dual triple J^1, J^2, J^3 ($\mathfrak{su}(2)$); the direct Fano tunnel B, which bypasses E_2, generates the commuting center ω ($\mathfrak{u}(1)$). The rank $\dim \mathcal{F}(p) = 4$ counts these channels.

Hooft tensor enters geometrically.

For aphanic p, the spectrum decomposition $\mathbb{S} = K_p \oplus E_2(p) \oplus E_4(p)$ — the eigenspaces of G_p for eigenvalues $0, \|p\|^2, 2\|p\|^2$ respectively — is preserved by every L_a with $a \in \mathfrak{q}_p$. For $a \notin \mathfrak{q}_p$, the operator L_a mixes the three eigenspaces, and the mixing is highly constrained.

Proposition 4.17 (Eigenspace Mixing Trichotomy)**.** *For aphanic p and $a \in \mathrm{Im}(\mathbb{S})$, decompose L_a into the nine $\mathrm{Hom}(E_\mu(p), E_\nu(p))$ blocks ($\mu, \nu \in \{0, 2, 4\}$, where $E_0(p) := K_p$). Define the off-diagonal block norms*

$$A_a := \pi_{K_p} L_a \, \pi_{E_2(p)}, \qquad B_a := \pi_{K_p} L_a \, \pi_{E_4(p)}, \qquad C_a := \pi_{E_2(p)} L_a \, \pi_{E_4(p)}.$$

Then for every $a \in \mathrm{Im}(\mathbb{S})$,

$$\|[L_a, L_p^2]\|_F^2 \;=\; \|p\|^4 \left(2(\|A_a\|_F^2 + \|C_a\|_F^2) \;+\; 8\,\|B_a\|_F^2 \right). \tag{34}$$

At the basepoint p_0, the imaginary basis directions partition into three classes:

Class	*Indices*	$\|A\|_F^2$	$\|B\|_F^2$	$\|C\|_F^2$	$\|[L_a, L_{p_0}^2]\|_F^2$
Centralizer $\mathfrak{q}_{p_0}$	$\{1, 2, 8, 9, 10\}$	0	0	0	0
Adjacent mixer	$\{4, 5, 6, 7, 12, 13, 14, 15\}$	4	0	4	64
Tunneler (Fano)	$\{3, 11\}$	0	4	0	128

The same table holds for any aphanic p after relabeling $\mathfrak{q}_p$ and rescaling by $\|p\|^4$.

Proof. Identity (34) is a direct computation: L_p^2 is self-adjoint with eigenvalues $\{0, -\|p\|^2, -2\|p\|^2\}$ on $\{K_p, E_2, E_4\}$ (recall $G_p = -L_p^2$ for imaginary p); the squared Frobenius norm of

$[L_a, L_p^2]$ decomposes block-by-block, and only off-diagonal blocks contribute, weighted by the squared eigenvalue gaps:

$$\|p\|^4\,(2\|A\|^2 + 2\|C\|^2) \;+\; 2\,(2\|p\|^2)^2\,\|B\|^2 = \|p\|^4(2\|A\|^2 + 8\|B\|^2 + 2\|C\|^2).$$

The basepoint table is verified by enumeration of $(\|A_{e_i}\|_F, \|B_{e_i}\|_F, \|C_{e_i}\|_F)$ for $i = 1, \ldots, 15$ (the probe `op_F1_quantization_block_structure.cpp` in the supplementary archive), and all 15 rows are certified in *exact integer arithmetic* by `op_six_anchors_B`: the three block projectors are exact integer polynomials in G_p, certified idempotent with traces $4, 8, 4$, and every rescaling is an exact division. The weights $(2, 8, 2)$ in the identity above are confirmed exactly, row by row. Universality across the aphanic locus follows from G_2-equivariance applied to each block projection (Lemma 4.6). □

Remark 4.18 (Fano elements as direct $K \leftrightarrow E_4$ tunnels). The Fano completions $e_3 = e_1 e_2$ and $e_{11} = e_3\,\ell$ at p_0 are the unique imaginary basis directions with $A = C = 0$ and $B \neq 0$ — they mix the kernel directly with the highest eigenspace, bypassing the intermediate E_2. This is the structural reason that the $\mathfrak{u}(1)$ generator $\omega = \mathcal{C}_{e_3, e_{11}}(p_0)$ exists as a fourth independent operator in $\mathcal{F}(p_0)$: the $K \leftrightarrow E_4$ tunneling produces a curvature contribution that the adjacent mixers cannot generate.

The proposition's value is conceptual rather than technical: Theorem 4.13 can be read off from Proposition 4.17 as the statement that the rank of $\mathcal{F}(p)$ equals the number of distinct mixing channels into K_p (three from adjacent mixing, one from Fano tunneling), exactly 4.

4.9 Summary

The closed-form representation of the curvature operator (Definition 4.4, Proposition 4.2) reduces the rank theorem to a finite-dimensional commutator question. The G_2-equivariance lemma (Lemma 4.6) reduces it further from the 11-dimensional aphanic locus to a single basepoint. The Q-Centralizer Vanishing Theorem (Theorem 4.10) exhausts the basepoint count by identifying the 5-dimensional kill set $\mathfrak{q}_p$ explicitly. The remaining 10-dimensional contribution splits, by the Eigenspace Mixing Trichotomy (Proposition 4.17), into eight adjacent mixers and two Fano tunnelers; the self-duality identities (28) and the octonion-half doubling (29) reduce these ten contributions to four linearly independent generators $\{J^1, J^2, J^3, \omega\}$.

The $\mathfrak{u}(1) \oplus \mathfrak{su}(2) = \mathfrak{u}(2)$ holonomy algebra of the aphanic bundle (curvature span by Theorem 4.13, capped to the holonomy algebra by Proposition 4.15) thus emerges from sedenion algebra alone, without any input from differential geometry or physics.

5 Discrete Formulas

Section 4 described the curvature of the aphanic bundle as an *operator*: the $\mathfrak{u}(2)$-valued Berry curvature $\mathcal{C}_{a,b}(p)$ on the kernel fibre. Monitoring a running system instead calls for a *scalar*: a single number per step measuring how fast the hidden sector turns along an actual trajectory. We now introduce that pathwise creative-curvature scalar κ_γ, the operational projection of the operator-level structure of §4 onto a sampled path rather than a new object. For computational implementation this continuous pathwise curvature must be discretized; we provide two equivalent forms.

5.1 Turn-Angle Form

Let $\{a_k\}_{k=0}^{N}$ be a discrete sequence of hidden-sector representatives along a sampled trajectory, normalized so that $\|a_k\| = 1$. Define:

Definition 5.1 (Discrete Creative Curvature)**.**

$$\kappa_\gamma[k] = \arccos(\langle a_k,\, a_{k+1}\rangle), \tag{35}$$

the geodesic angle between successive hidden generators on the unit sphere.

This is the discrete analogue of equation (19): angular displacement per step. When a_k and a_{k+1} are nearly parallel, $\kappa_\gamma[k] \approx 0$ (refinement). When they are nearly orthogonal, $\kappa_\gamma[k] \approx \pi/2$ (maximal turning).

Gauge invariance. Each a_k is a unit representative chosen within the 4-dimensional fiber K_{p_k}, so $\langle a_k,\, a_{k+1}\rangle$, and hence $\kappa_\gamma[k]$ as written, depends on that choice of section, i.e. it is gauge-dependent. The section-independent quantity is the principal angle between the kernel subspaces K_{p_k} and $K_{p_{k+1}}$, equivalently the projector distance $\|\pi_{k+1} - \pi_k\|_F$ (with π_k the closed-form projector of Proposition 4.2); (35) recovers the leading principal angle once a parallel-transported section is fixed. We use the projector-distance form wherever a gauge-invariant diagnostic is required.

5.2 G-Metric Form

A second form weights the inter-fiber step by the ambient Gram operator $G_k = L_{p_k}^T L_{p_k}$. We emphasize that G_k is *not* a metric on the hidden sector itself: K_{p_k} is exactly the zero eigenspace of G_k, so $G_k|_{K_{p_k}} = 0$ (the 4–8–4 spectrum). The chord $\delta a_k = a_{k+1} - a_k$, however, leaves K_{p_k}, and

$$\kappa_\gamma[k] = \sqrt{\delta a_k^T\, G_k\, \delta a_k}, \quad \delta a_k = a_{k+1} - a_k, \tag{36}$$

reduces, using $G_k a_k = 0$, to $\kappa_\gamma[k] = \sqrt{a_{k+1}^T G_k\, a_{k+1}}$: the energy the current Gram operator assigns to the incoming generator. It vanishes iff $a_{k+1} \in K_{p_k}$ (the kernel has

not moved) and is non-degenerate precisely because it probes the chord *between* fibers, not a metric within one.

5.3 Exogenous Subtraction

Not all hidden-sector rotation is internally generated. External perturbations (new data, user interventions, environmental changes) can rotate a_k without the system doing any creative work. We subtract these.

Definition 5.2 (Exogenous Subtraction)**.** Let E_k be the subspace of hidden-sector variation attributable to exogenous input at step k. The *intrinsic* creative curvature is:

$$\kappa_\gamma^{\text{int}}[k] = \left\| (\Pi_{E_k^\perp} \delta a_k) \right\|, \tag{37}$$

where $\Pi_{E_k^\perp}$ is orthogonal projection onto the complement of E_k.

Proposition 5.3 (Graceful Degradation)**.** *If E_k cannot be fully identified (rank-deficient estimation of the exogenous subspace), then $\kappa_\gamma^{\text{int}}[k]$ overestimates intrinsic curvature by at most $\|\Pi_{E_k} \delta a_k\|$, and in particular:*

$$\kappa_\gamma^{\text{int}}[k] \leq \kappa_\gamma[k].$$

The estimate is conservative: unidentified exogenous components are attributed to intrinsic creativity, not the reverse.

Proof. By orthogonal decomposition, $\delta a_k = \Pi_{E_k^\perp} \delta a_k + \Pi_{E_k} \delta a_k$. Then $\left\| \Pi_{E_k^\perp} \delta a_k \right\| \leq \|\delta a_k\| = \kappa_\gamma[k]$. Equality holds when $E_k = \{0\}$, i.e., when no exogenous component is subtracted. The overestimate $\kappa_\gamma^{\text{int}} - \kappa_\gamma^{\text{true}} = \left\| \Pi_{\tilde{E}_k^\perp \cap E_k} \delta a_k \right\|$ vanishes when $\tilde{E}_k \supseteq E_k$. □

5.4 Phase Transition Score

Definition 5.4 (Phase Transition Score)**.**

$$\Phi_k = \varrho_k \cdot \kappa_\gamma[k], \tag{38}$$

where ϱ_k is the *mass* of state k, a measure of the state's influence on downstream computation (e.g., attention weight, probability mass, or allocation fraction).

Mass-weighted curvature serves as a cheap pre-filter: high Φ_k flags states where a massive (influential) component is turning rapidly, the signature of a phase transition. Low-mass components can turn freely without systemic consequence.

6 The State Vector

At each step k, the curvature monitor observes the composite state vector:

$$x_k = (\, a_k,\ \kappa_\gamma[k],\ V_k,\ \varrho_k,\ \sigma_k^{\text{surv}},\ C_k^{\text{surv}}\,), \tag{39}$$

where $a_k \in \ker(L_{\gamma(k)})$ is the hidden-sector representative, $\kappa_\gamma[k]$ is the discrete curvature, V_k is the verification weight, ϱ_k is the mass, and $\sigma_k^{\text{surv}} \in \{0, 1\}$ is the survival indicator (does the creative output persist beyond a confirmation horizon L_{confirm}?).

The hidden projection $\Pi_{\mathcal{A}} : V \to \ker(L_p)$ extracts the component of any vector that lies in the hidden sector. When a metric G is available on the ambient space V, the Gram structure $G|_{\ker(L_p)}$ provides the pullback metric on the hidden sector.

7 Verified Creativity

Curvature alone does not distinguish creation from hallucination. A system that turns rapidly while breaking every contract is not creative; it is unmoored. Verification provides the second leg.

Definition 7.1 (Verification Weight)**.** The *verification weight* $V_k \in [0, 1]$ is the fraction of the curvature at step k that survives dialectical verification. In a thesis-antithesis-synthesis architecture:

$$V_k = \frac{\left\| a_k^{\text{post-synthesis}} \right\|}{\left\| a_k^{\text{post-thesis}} \right\|},$$

the ratio of hidden-sector magnitude after synthesis to its magnitude after thesis.

Definition 7.2 (Speed Gate)**.** The *speed gate* $\sigma_k^{\text{speed}} \in \{0, 1\}$ prevents trivially slow trajectories from registering as creative:

$$\sigma_k^{\text{speed}} = \begin{cases} 1 & \text{if } \|\dot{\gamma}_k\| > v_{\min}, \\ 0 & \text{otherwise.} \end{cases} \tag{40}$$

Definition 7.3 (Survival-Gated Creativity Score)**.** The *survival-gated verified creativity score* is:

$$C_k^{\text{surv}} = \kappa_\gamma[k] \cdot V_k \cdot \sigma_k^{\text{surv}}. \tag{41}$$

The triple product gates on three independent conditions: the system must be turning ($\kappa_\gamma > 0$), the turn must survive verification ($V_k > 0$), and the result must persist beyond the confirmation horizon ($\sigma_k^{\text{surv}} = 1$). Failure of any single factor zeroes the score.

Remark 7.4 (Failure Mode Separation)*.* The score C_k^{surv} admits exactly five failure modes, each diagnosable from a distinct component of the triple product (Table 1). Since C_k^{surv} is a product of three non-negative factors, it vanishes if and only if at least one factor vanishes. The first factor κ_γ vanishes in stagnation. The second V_k is low in hallucination

and medium in drift. The third σ^{surv} vanishes in ephemeral sparks. The five modes are exhaustive under the decomposition $C_k^{\text{surv}} > 0$ vs. $C_k^{\text{surv}} = 0$ crossed with which factor is responsible.

Table 1: Failure mode taxonomy for C_k^{surv}.

Mode	κ_γ	V_k	σ^{surv}	**Diagnosis**
Stagnation	≈ 0	–	–	Hidden sector stationary; no structural change
Hallucination	> 0	low	–	Turning fails verification; unmoored from constraints
Ephemeral spark	> 0	high	0	Verified turn that does not persist
Drift	> 0	med	1	Persistent but weakly verified; slow quality erosion
Genuine creation	> 0	high	1	Verified, persistent structural transformation

8 The Seven-Regime Taxonomy

The continuous quantities $(\kappa_\gamma, V_k, \Phi_k, C_k^{\text{surv}})$ are mapped to seven discrete operational regimes via threshold classification.

Table 2: Seven-regime operational taxonomy.

Regime	κ_γ	V_k	Φ_k	C_k^{surv}	**System state**
1. Dormant	≈ 0	–	low	0	No hidden-sector activity
2. Refining	low	high	low	≈ 0	Smooth exploitation within constraints
3. Exploring	med	med	med	low	Testing constraint boundaries
4. Creating	high	high	high	high	Verified deformation of constraint manifold
5. Drifting	med	low	med	low	Gradual departure from constraints
6. Hallucin.	high	low	high	0	Rapid unconstrained departure
7. Transition	spike	–	spike	–	Discontinuous hidden-sector restructuring

Remark 8.1 (Pivot Detection). The transition Regime 4 $\to$ Regime 6 $\to$ Regime 4 (creation followed by brief hallucination followed by renewed creation) is a *pivot*: the system temporarily loses verification while restructuring, then regains it at a new constraint level. Pivots are desirable. The hysteresis parameter L_{confirm} prevents the monitor from penalizing pivots by requiring that regime transitions persist for at least L_{confirm} steps before triggering operational responses.

Remark 8.2 (Silent Decay). The transition Regime 2 $\to$ Regime 5, from refining to drifting, is the most dangerous failure mode because it involves no curvature spike. The system does not suddenly hallucinate; it gradually loses verification quality while continuing to produce apparently normal output. This is the regime where V_k tracking is essential and κ_γ alone is insufficient.

Remark 8.3 (Curvature Without Verification). The verification weight V_k presupposes a verification oracle (see Limitation 4 in Section 10.3). However, the framework retains value even when V_k is unavailable. Setting $V_k = 1$ (no verification), the curvature κ_γ alone distinguishes stasis from motion, and the holonomy phase $\theta(\Gamma)$ classifies closed loops independently of verification. The contribution of this paper is the geometric monitoring layer; the verification interface is modular.

9 A Holonomy Phase for Closed Arcs

We now define a U(1) *holonomy phase* that classifies closed dialectical arcs by the net structural turning they accumulate. We stress at the outset that this is a *geometric* (holonomy) invariant, not a topological one: the aphanic base is simply connected (it is an $\mathbb{R}^+$-cone over $G_2/\mathrm{SU}(2)$, and $\pi_1(G_2/\mathrm{SU}(2)) = 0$), so the phase is curvature-driven and deformation-sensitive, not quantized by homotopy.

Let $\Gamma : [0,1] \to \mathcal{P}_{\mathrm{aph}}$ be a closed loop in the aphanic locus ($\Gamma(0) = \Gamma(1)$). Along Γ the hidden sector $\ker(L_{\Gamma(t)})$ varies smoothly. The real rank-4 kernel bundle carries the ∇-parallel complex structure $J = R_\ell|_K$ of Proposition 4.15, which makes each fiber a complex space and supplies a genuine U(1) connection 1-form: for a unit section a of K,

$$\alpha(X) \;:=\; \langle J\,a,\ \nabla_X a\rangle. \tag{42}$$

The naive real expression $\langle a, \nabla_X a\rangle = \frac{1}{2}X\|a\|^2$ vanishes identically for a unit section; the complex structure J is exactly what makes the abelian phase nonzero, playing the role of the factor i in the Hermitian Berry connection.

Definition 9.1 (Holonomy phase)**.** The *holonomy phase* of a closed dialectical arc Γ is

$$\theta(\Gamma) \;=\; \oint_\Gamma \alpha \;\in\; \mathbb{R}/2\pi\mathbb{Z} \;\cong\; \mathrm{U}(1), \tag{43}$$

the U(1) holonomy of the connection (42) around Γ.

The phase $\theta(\Gamma)$ is *not* integer-valued or homotopy-protected. Because $\mathcal{P}_{\mathrm{aph}}$ is simply connected, every loop is contractible and $\theta(\Gamma)$ varies continuously under deformation of Γ: by Stokes it equals the enclosed abelian curvature flux, $\theta(\Gamma) = \iint_D F^\nabla\big|_{\mathfrak{u}(1)}$ for any disk D with $\partial D = \Gamma$. It is the abelian U(1) component of the bundle's holonomy, whose full holonomy algebra is $\mathfrak{u}(2)$ (Proposition 4.15); the non-abelian SU(2) component is the Wilczek–Zee generalization and is not used here. The kernel bundle *does* carry a genuine topological invariant, a nontrivial $\mathbb{Z}/2$ characteristic class, the "aphanic twist", but it lives in $H^6(G_2/\mathrm{SU}(2); \mathbb{Z}/2)$ [21] and classifies the *bundle*, not individual arcs: it is detected by five- and six-dimensional cycles, never by π_1, and is therefore unrelated to $\theta(\Gamma)$.

Proposition 9.2 (Classification of Dialectical Arcs)**.** *Let Γ be a closed loop in the aphanic locus $\mathcal{P}_{\text{aph}}$, and let $\bar{\kappa_\gamma}(\Gamma) = \frac{1}{|\Gamma|}\oint_\Gamma \kappa_\gamma(t)\,dt$ be the mean creative curvature. Then exactly one of the following holds:*

(i) *$\bar{\kappa_\gamma} = 0$ and $\theta(\Gamma) = 0$: **Stasis loop.** The hidden sector is stationary and the loop produces no structural change.*

(ii) *$\bar{\kappa_\gamma} > 0$ and $\theta(\Gamma) = 0$: **Drift loop.** The hidden sector rotates but returns to its initial configuration. The system explores without transforming.*

(iii) *$\bar{\kappa_\gamma} > 0$ and $\theta(\Gamma) \neq 0$: **Turning loop.** The hidden sector accumulates a nonzero net holonomy phase and does not return to its initial orientation. The arc has enclosed nonzero curvature flux, a quantitative certificate of net structural turning. (This is geometric, not homotopy-protected; see the proof.)*

The mathematical content is concentrated in case (i)*: vanishing curvature implies trivial holonomy. Cases* (ii) *and* (iii) *are definitional: they organize the operational taxonomy by partitioning the $(\bar{\kappa_\gamma}, \theta(\Gamma))$ plane.*

Proof. **Case (i).** If $\bar{\kappa_\gamma} = 0$, then $\kappa_\gamma(t) = 0$ for almost all t (since $\kappa_\gamma \geq 0$). The hidden generator is parallel-transported: $\nabla_{\dot\gamma} a = 0$ along Γ. The connection form $\alpha = \langle Ja, \nabla_{\dot\gamma} a\rangle$ vanishes along Γ, so $\theta(\Gamma) = \oint_\Gamma 0\,dt = 0$.

Case (ii). $\bar{\kappa_\gamma} > 0$ implies the hidden sector rotates along Γ, but $\theta(\Gamma) = 0$ in $\mathbb{R}/2\pi\mathbb{Z}$ means the net accumulated phase is trivial: the hidden sector returns to its initial orientation despite moving.

Case (iii). $\theta(\Gamma) \neq 0$ means the U(1) holonomy around Γ is nontrivial: the hidden sector does not return to its initial orientation in the complex line spanned by $\{a, Ja\}$. By Stokes the arc encloses nonzero abelian curvature flux, $\theta(\Gamma) = \iint_D F^\nabla|_{\mathfrak{u}(1)} \neq 0$. This is a *quantitative* certificate of net turning, not a discrete topological charge, and it is not homotopy-protected: since $\mathcal{P}_{\text{aph}}$ is simply connected, Γ can be contracted and $\theta(\Gamma)$ driven continuously to zero as the enclosed flux shrinks. The phase records how much curvature the arc actually swept. □

Remark 9.3 (The Regime Sequence as Holonomy Instrument). The regime taxonomy of Section 8 provides the *local* diagnostic (what is the system doing now?). The holonomy phase provides the *global* diagnostic (what did the system accomplish over a complete cycle?). Together they form a complete instrument: the regime sequence is the trajectory, and $\theta(\Gamma)$ is its holonomy shadow.

10 Discussion

10.1 Relationship to Information Geometry

Information geometry and aphanics operate on complementary structures. The Fisher metric measures curvature of the *visible* statistical manifold: how predictions change as

parameters vary. Creative curvature measures curvature of the *hidden* sector: how the kernel of the action operator rotates as the system moves. In a system where the Fisher metric is flat (parameters change smoothly without surprises), the creative curvature can be large (the hidden sector restructures dramatically). The two metrics are generically independent.

A deeper connection may exist through the null space of the Fisher matrix itself: directions in parameter space along which the likelihood surface is flat correspond to structural unidentifiability: parameters the data cannot see. These are precisely the "hidden generators" from the statistical perspective. Whether the null space of the Fisher matrix carries the same bundle structure as $\ker(L_p)$ is an open question.

10.2 The Null-Space Inversion

The central philosophical move of aphanics, treating the null space as generative rather than deficient, has a concrete mathematical consequence. In standard treatments, $\ker(L_p) \neq \{0\}$ signals a *singularity*: the operator is non-invertible, the system is degenerate, the computation fails. Aphanics reinterprets this singularity as the locus of maximal structural richness. The hidden sector is largest precisely where the system is most "degenerate" in the classical sense, and this is where creative curvature is most interesting.

The motivating algebraic setting sharpens this picture: the abelian component of the holonomy on the hidden sector is generated by an ambient operator on the parent algebra (Section 4.1), and the same identification recurs at every Cayley–Dickson level where the hidden sector exists. The null-space inversion is therefore not opportunistic relabeling but a structural discovery: the kernel inherits a measurement operator from the algebra it sits inside, and that inherited operator (the complex structure $J = R_\ell|_K$) is what supplies the abelian U(1) holonomy phase in case (iii) of Proposition 9.2.

This inversion has precedent. Zero divisors in non-associative algebras were long considered pathological; recent work has shown them to be algebraically and geometrically rich. Singularities in algebraic geometry are the loci of maximal information density. Phase transitions in statistical mechanics occur at parameter values where the partition function is singular. In each case, the null space is not empty space; it is compressed structure.

Remark 10.1 (Cross-Domain Resonances). The sign-flip from deficiency to generativity has structural parallels across domains: Gödel's incompleteness (the unprovable statement is what prevents collapse into decidability), Heisenberg's uncertainty (the non-commuting observable is what allows conjugate properties to coexist), and Arrow's impossibility (the intransitive social preference is what prevents dictatorial aggregation). We do not claim these are instances of a single mathematical theorem; the structures differ in essential ways. We note only that the *interpretive move* of treating a null space as generative rather than deficient recurs across formalisms, and that aphanics provides one concrete

algebraic setting where this move has precise content.

10.3 Limitations

We note six limitations of the present framework:

1. **Theorem/proposal boundary.** Sections 3–4 establish geometric *theorems* in the sedenion model. Sections 5–9 define an instrumentation *template* for dialectical systems; applying it to an actual deployed system requires that the system expose, or approximate, a hidden-sector bundle with the stated structure. That the templates apply to any particular running system is not established here.

2. **Geometric, not topological, loop phase.** The aphanic base considered here is simply connected, so the holonomy phase $\theta(\Gamma)$ is *not* a homotopy invariant of loops. It is a curvature-driven holonomy phase: deformation-sensitive, quantitative, and useful as a certificate of net structural turning, but not quantized by π_1. The only topologically protected quantity is the $\mathbb{Z}/2$ class of the bundle itself (in H^6), which classifies the bundle, not individual arcs.

3. **Connection dependence.** The holonomy phase $\theta(\Gamma)$ depends on the choice of connection. The framework uses the canonical kernel-projector connection together with the parallel complex structure $J = R_\ell|_K$ (Proposition 4.15); this is natural in the sedenion setting but a different bundle would require its own canonical choice.

4. **Exogenous estimation.** The exogenous subtraction procedure (Section 5.3) requires identifying the exogenous subspace E_k. In practice this is approximate, and Proposition 5.3 only guarantees conservative (upward) bias. Systematic overestimation of intrinsic curvature could lead to false-positive creativity detection.

5. **Verification oracle.** The verification weight V_k presupposes access to a verification procedure. In dialectical architectures this is built in (the antithesis). In other architectures it must be supplied externally, and the quality of V_k depends on the quality of the oracle.

6. **Computational cost.** Computing κ_γ requires tracking the kernel of L_p at every step, which may be expensive in high-dimensional systems. The discrete turn-angle form (Section 5.1) is $O(\dim V)$ per step, but the G-metric form (Section 5.2) requires the full Gram operator.

11 Conclusion

We have introduced aphanics, the study of structurally real generators that are annihilated by a system's own internal action, and derived from it a geometric framework for

monitoring creativity in cognitive and dialectical systems. The framework rests on three pillars:

Creative curvature κ_γ measures the rate at which the hidden sector rotates orthogonal to the system trajectory. It is intrinsic, parameterization-free, and distinguishes refinement from genuine structural turning.

Verified creativity C_k^{surv} gates curvature on dialectical verification and temporal persistence, separating five failure modes (stagnation, hallucination, ephemeral spark, drift, genuine creation) from a single observable triple product.

The holonomy phase $\theta(\Gamma)$ classifies closed dialectical arcs by the net curvature flux they enclose, providing a U(1) geometric (holonomy) certificate (deformation-sensitive rather than topological, since the aphanic base is simply connected) that distinguishes turning loops from drift.

The framework is constructive: we have provided discrete formulas, operational taxonomies, and a failure-mode classification that can be instrumented in any system with access to its hidden sector. The algebraic foundations that validate these tools in a concrete setting, the sedenion algebra, are reported in a companion work [21]. The present framework is a proposal; its ultimate value depends on the depth of that algebraic foundation and on empirical demonstration in operational systems.

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This paper is dedicated to Karla, who has struggled the most in understanding and given the most in trying, and to Aristotle and Louise, to show them what being is, how to live: that we are not ridiculous; that "a lot" is often exactly what is needed; and that when the way things are is not the way they should be, and the way they should be is not the way they are, it is your responsibility to do something about it, or lose yourself to it.

A Expanded Proofs

A.1 Proof of Proposition 5.3 (Graceful Degradation)

Proof. Let E_k be the true exogenous subspace and $\tilde{E}_k$ the estimated subspace. Write $\delta a_k = \delta a_k^{\text{int}} + \delta a_k^{\text{exo}}$ where $\delta a_k^{\text{int}} \in E_k^{\perp}$ and $\delta a_k^{\text{exo}} \in E_k$.

The estimated intrinsic component is $\Pi_{\tilde{E}_k^{\perp}} \delta a_k$. If $\tilde{E}_k \subseteq E_k$ (underestimation of the exogenous subspace), then $E_k^{\perp} \subseteq \tilde{E}_k^{\perp}$, so $\Pi_{\tilde{E}_k^{\perp}}$ retains all of δa_k^{int} plus some of δa_k^{exo}:

$$\left\| \Pi_{\tilde{E}_k^{\perp}} \delta a_k \right\| \geq \left\| \delta a_k^{\text{int}} \right\| = \kappa_\gamma^{\text{true}}[k].$$

The overestimate is $\left\| \Pi_{\tilde{E}_k^{\perp}} \delta a_k^{\text{exo}} \right\| \leq \|\delta a_k^{\text{exo}}\|$. Since $\|\delta a_k^{\text{int}}\|^2 + \|\delta a_k^{\text{exo}}\|^2 = \|\delta a_k\|^2 = \kappa_\gamma[k]^2$, we have $\kappa_\gamma^{\text{int}}[k] \leq \kappa_\gamma[k]$. □

A.2 Proof of Proposition 9.2 (Classification of Dialectical Arcs)

Proof. We expand the three cases from the main text. Throughout, $\alpha(X) = \langle Ja, \nabla_X a \rangle$ is the U(1) connection (42) built from the parallel complex structure $J = R_\ell|_K$ (Proposition 4.15); the naive real form $\langle a, \nabla_X a \rangle = \frac{1}{2} X \|a\|^2$ vanishes identically and carries no phase, which is precisely why J is needed.

Case (i): $\bar{\kappa_\gamma} = 0$ implies $\theta(\Gamma) = 0$. Since $\kappa_\gamma(t) \geq 0$ and $\bar{\kappa_\gamma} = \frac{1}{|\Gamma|} \oint \kappa_\gamma \, dt = 0$, we have $\kappa_\gamma(t) = 0$ a.e., so $\nabla_{\dot{\gamma}} a = 0$ a.e.: a is parallel along Γ. Then $\alpha = \langle Ja, \nabla_{\dot{\gamma}} a \rangle = 0$ a.e., giving $\theta(\Gamma) = 0$.

Case (ii): $\bar{\kappa_\gamma} > 0$, $\theta(\Gamma) = 0$. The hidden sector rotates but the net phase $\oint \alpha = 0$ in $\mathbb{R}/2\pi\mathbb{Z}$. The U(1) holonomy is trivial: a returns to its initial orientation in the complex line $\text{span}\{a, Ja\}$. The arc explores the hidden sector and returns: displacement without net turning.

Case (iii): $\bar{\kappa_\gamma} > 0$, $\theta(\Gamma) \neq 0$. The U(1) holonomy $\exp(i \oint \alpha) \neq 1$: the hidden generator does not return, $a(\Gamma(1))$ and $a(\Gamma(0))$ differing by the nonzero rotation angle $\theta(\Gamma)$ in $\text{span}\{a, Ja\}$. By Stokes, $\theta(\Gamma) = \iint_D F^{\nabla}|_{\mathfrak{u}(1)}$ for any disk D bounding Γ, so a nonzero phase certifies nonzero enclosed curvature flux. Because $\mathcal{P}_{\text{aph}}$ is simply connected ($\pi_1 = 0$), this is *not* homotopy-protected: Γ can be contracted and $\theta(\Gamma)$ driven continuously to 0, the flux through the shrinking disk vanishing. The phase is a quantitative record of the curvature the arc swept, not a discrete topological charge. □

B Conceptual Monitor: Pseudocode

Algorithm 1 summarizes the curvature monitor as an eight-step procedure. No implementation-specific details (data structures, memory layout, SIMD kernels, or language-specific constructs) are exposed.

The monitor is *passive*: it observes the trajectory and classifies, but does not intervene. Intervention policies (throttle, halt, rollback) are downstream of classification and outside

Algorithm 1: Curvature Monitor (Conceptual)

Input: Trajectory γ, action operator L, exogenous estimator $\hat{E}$, verification oracle $\mathcal{V}$, thresholds $(\tau_\kappa, \tau_V, v_{\min}, L_{\text{confirm}})$

Output: Regime sequence $R_0, R_1, \ldots$

for *each step* k **do**

- **Step 1:** Compute $a_k \leftarrow$ unit vector in $\ker(L_{\gamma(k)})$;
- **Step 2:** Compute raw curvature $\kappa_\gamma[k] \leftarrow \arccos(\langle a_{k-1}, a_k \rangle)$;
- **Step 3:** Estimate exogenous subspace $E_k \leftarrow \hat{E}(k)$;
- **Step 4:** Subtract exogenous component: $\kappa_\gamma^{\text{int}}[k] \leftarrow \left\| \Pi_{E_k^\perp}(a_k - a_{k-1}) \right\|$;
- **Step 5:** Compute verification weight $V_k \leftarrow \mathcal{V}(a_k^{\text{thesis}}, a_k^{\text{synth}})$;
- **Step 6:** Compute phase transition score $\Phi_k \leftarrow \varrho_k \cdot \kappa_\gamma^{\text{int}}[k]$;
- **Step 7:** Compute survival-gated score $C_k^{\text{surv}} \leftarrow \kappa_\gamma^{\text{int}}[k] \cdot V_k \cdot \sigma_k^{\text{surv}}$;
- **Step 8:** Classify regime $R_k \leftarrow \text{Classify}(\kappa_\gamma^{\text{int}}, V_k, \Phi_k, C_k^{\text{surv}}; \tau_\kappa, \tau_V)$ with L_{confirm}-step hysteresis;

end

the scope of this paper.

C The Basepoint Table

This appendix exhibits the explicit data underlying the proof of Theorem 4.13. All entries are computed at $p_0 = e_1 + e_{10}$ and verified at machine precision (maximum residual $0e+00$).

C.1 Canonical kernel basis

The Gram–Schmidt-orthonormalized output of `compute_kernel` [21] at p_0 yields

$$\begin{aligned} v_0 &= \tfrac{1}{\sqrt{2}}(e_7 + e_{12}), \quad v_1 = \tfrac{1}{\sqrt{2}}(-e_6 + e_{13}), \\ v_2 &= \tfrac{1}{\sqrt{2}}(e_5 + e_{14}), \quad v_3 = \tfrac{1}{\sqrt{2}}(-e_4 + e_{15}). \end{aligned} \tag{44}$$

These four vectors form an orthonormal basis of K_{p_0}; $L_{p_0} v_k = 0$ for $k = 0, 1, 2, 3$.

C.2 The four generators in the kernel basis

In the $\{v_0, v_1, v_2, v_3\}$ basis, the four generators of $\mathcal{F}(p_0)$ defined in (30) have the explicit matrix forms

$$J^1 = \begin{pmatrix} 0 & -\frac{1}{2} & 0 & 0 \\ \frac{1}{2} & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{1}{2} \\ 0 & 0 & \frac{1}{2} & 0 \end{pmatrix}, \qquad J^2 = \begin{pmatrix} 0 & 0 & -\frac{1}{2} & 0 \\ 0 & 0 & 0 & \frac{1}{2} \\ \frac{1}{2} & 0 & 0 & 0 \\ 0 & -\frac{1}{2} & 0 & 0 \end{pmatrix}, \tag{45}$$

$$J^3 = \begin{pmatrix} 0 & 0 & 0 & -\frac{1}{2} \\ 0 & 0 & -\frac{1}{2} & 0 \\ 0 & \frac{1}{2} & 0 & 0 \\ \frac{1}{2} & 0 & 0 & 0 \end{pmatrix}, \qquad \omega = \begin{pmatrix} 0 & 0 & 0 & -\frac{1}{2} \\ 0 & 0 & \frac{1}{2} & 0 \\ 0 & -\frac{1}{2} & 0 & 0 \\ \frac{1}{2} & 0 & 0 & 0 \end{pmatrix}. \tag{46}$$

Each is skew-symmetric of Frobenius norm one. The triple $\{J^1, J^2, J^3\}$ exhibits the standard $\mathfrak{su}(2)_+$ action on $\mathbb{R}^4$ (self-dual sector of $\mathfrak{so}(4)$), and ω is skew-symmetric with $(2\omega)^2 = -I$: that is, 2ω is a *complex structure* on K_{p_0} (not an involution: a nonzero real skew matrix satisfies $\omega^2 \preceq 0$, so $\omega^2 = +I$ is impossible). Up to sign it is the restriction $R_\ell|_{K_{p_0}}$ of right multiplication by the Cayley–Dickson doubling generator (Proposition 4.15), and it commutes with the triple.

C.3 Decomposition of the nonzero curvature pairs

All 25 nonzero curvature operators $\mathcal{C}_{e_i,e_j}(p_0)$ with $1 \le i < j \le 15$ admit unique integer decompositions

$$\mathcal{C}_{e_i,e_j}(p_0) \;=\; c_1\, J^1 + c_2\, J^2 + c_3\, J^3 + c_\omega\, \omega, \qquad (c_1, c_2, c_3, c_\omega) \in \mathbb{Z}^4.$$

Group I — within first octonion half ($i, j \in \{4, 5, 6, 7\}$):

(i, j)	c_1	c_2	c_3	c_ω	$\|\mathcal{C}\|_F$
(e_4, e_5)	-2	+0	+0	+0	2.000
(e_4, e_6)	+0	-2	+0	+0	2.000
(e_4, e_7)	+0	+0	-2	+0	2.000
(e_5, e_6)	+0	+0	-2	+0	2.000
(e_5, e_7)	+0	+2	+0	+0	2.000
(e_6, e_7)	-2	+0	+0	+0	2.000

Group II — within second octonion half ($i, j \in \{12, 13, 14, 15\}$):

(i, j)	c_1	c_2	c_3	c_ω	$\|\mathcal{C}\|_F$
(e_{12}, e_{13})	-2	+0	+0	+0	2.000
(e_{12}, e_{14})	+0	-2	+0	+0	2.000
(e_{12}, e_{15})	+0	+0	-2	+0	2.000
(e_{13}, e_{14})	+0	+0	-2	+0	2.000
(e_{13}, e_{15})	+0	+2	+0	+0	2.000
(e_{14}, e_{15})	-2	+0	+0	+0	2.000

Group III — cross-half adjacent mixers ($i \in \{4, 5, 6, 7\},\ j \in \{12, 13, 14, 15\}$):

(i,j)	c_1	c_2	c_3	c_ω	$\|\mathcal{C}\|_F$
(e_4, e_{12})	+0	+0	-1	+1	1.414
(e_4, e_{13})	+0	+1	+0	+0	1.000
(e_4, e_{14})	-1	+0	+0	+0	1.000
(e_5, e_{12})	+0	+1	+0	+0	1.000
(e_5, e_{13})	+0	+0	+1	+1	1.414
(e_5, e_{15})	-1	+0	+0	+0	1.000
(e_6, e_{12})	-1	+0	+0	+0	1.000
(e_6, e_{14})	+0	+0	+1	+1	1.414
(e_6, e_{15})	+0	-1	+0	+0	1.000
(e_7, e_{13})	-1	+0	+0	+0	1.000
(e_7, e_{14})	+0	-1	+0	+0	1.000
(e_7, e_{15})	+0	+0	-1	+1	1.414

Group IV — Fano cross-half (e_3, e_{11}):

(i,j)	c_1	c_2	c_3	c_ω	$\|\mathcal{C}\|_F$
(e_3, e_{11})	+0	+0	+0	+1	1.000

C.4 Spectrum cross-check

From the table above, the 15×15 Gram matrix $(\Gamma_{\mathfrak{c}})_{ij} = \sum_k \langle \mathcal{C}_{e_i,e_k}(p_0), \mathcal{C}_{e_j,e_k}(p_0) \rangle_F$ can be reconstructed in closed form: each entry is a quadratic form in the integer coefficients $(c_1, c_2, c_3, c_\omega)$ with the orthonormality relations $\langle J^a, J^b \rangle_F = \delta_{ab}$, $\langle J^a, \omega \rangle_F = 0$, $\langle \omega, \omega \rangle_F = 1$. Direct evaluation gives the spectrum

$$\mathrm{Spec}(\Gamma_{\mathfrak{c}}) \;=\; \{\, 0^5,\, 1^2,\, 4^4,\, 28^4 \,\}, \tag{47}$$

matching (25). The five zero eigenvalues correspond to the five-dimensional Q-centralizer $\mathfrak{q}_{p_0} = \mathrm{span}\{e_1, e_2, e_8, e_9, e_{10}\}$.

This completes the proof of Theorem 4.13: the table exhibits all 25 nonzero curvature operators as integer combinations of the four-element generating set $\{J^1, J^2, J^3, \omega\}$, and the linear independence of the generating set is witnessed directly by the matrix forms (45).

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